

## MASTER

### Following a Virtual Lead-Vehicle

### Examining Different World-Fixed AR HUD Concepts for Car Navigation

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*Award date:*  
2021

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Eindhoven, 01-07-2021



**Following a Virtual Lead-Vehicle: Examining  
Different World-Fixed AR HUD Concepts for  
Car Navigation.**

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in partial fulfilment of the requirements for the degree of

**Master of Science  
in Human-Technology Interaction**

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## ACKNOWLEDGEMENTS

Many times, while working on this master thesis, there have been big mountains to climb, asking the most of me, all contributing to reaching the top: presenting my master thesis, marking the end of my Master program in Human-Technology Interaction. Namely, besides that this project aims to contribute to the development of augmented reality head-up display concepts in the automotive industry, this project has been a very useful learning experience for me personally. First of all, I have had the opportunity to be an intern at the UX Research department of TomTom Amsterdam, which showed me how my HTI background can be useful in an industry context. Additionally, it made me look forward to pursuing a career in a team with like-minded professionals working on UX challenges, which is pretty awesome considering that the majority of our meetings have been via a video call because of the corona pandemic. I would like to thank the whole TomTom UX research team for giving me this opportunity and taking me on the team.

One of the challenges I set myself was to improve my coding skills. I am happy to tell that I mastered the basics of programming in C# which was required to create a virtual environment in Unity3D. Next to that, I learned to use pandas in Python, making it possible to pre-process big data files and transform them into a single data file ready for data analysis. In addition, I learned creating fancy data visualizations using the language R. To me, these are big accomplishments, that I can hopefully use for my future career.

Another challenge during my project was much more imposed on me than intended, which was the ability to work alone. The corona pandemic challenged this to the extreme, showing me how much more I prefer to work and discuss together with other people compared to solving problems on my own. However, it also taught me that with effort I am able to find solutions, keep working, and make the best out of it. Here specifically, I would like to thank my friends (Maxine, Simon, Julia and Esther) where I could work on my thesis when I needed distraction and later, when working at university was possible again, the fellow HTI students/friends (Karlijn and Dick) which who I could discuss master thesis related questions.

During my project I was happy to be guided by three supervisors. First of all, I would like to thank Ruben Wesselink, who I first contacted for an internship at TomTom and later became my external supervisor. Thanks for all your help concerning insights in the automotive domain, more specifically in-vehicle systems or navigation systems, but also for sharing both happy and difficult moments throughout the project. Next, I would like to thank Raymond Cuijpers, who has assisted me with weekly meetings which always took way longer than expected. You were always able to discuss the difficult questions I had. And finally, Bastian Pfleging deserves appreciation as he made it possible to use the driving simulator and provided me with relevant insights into the automotive research area.

Finally, I would like to thank my family, who I visited more often than ever before during my student life, simply because I know they are always there for me and because I developed corona hobbies that are similar to theirs: discovering new nature areas both by bike or foot, and learning more about the flora and fauna, specifically birds.

## ABSTRACT

In the automotive industry, Head-Up Display (HUD) concepts for navigation are becoming more common. Augmented Reality (AR) HUD navigation solutions offer the opportunity to map information directly onto the relevant real-world situation, making the abstract information easier to interpret (Charissis & Papanastasiou, 2010). One concept that might serve as a proper AR HUD navigation system is a virtual lead-vehicle. Research has already shown that the dynamic virtual lead-vehicle has better or equal navigation performance, mental workload, and driving performance compared to screen-fixed AR HUD alternatives, a traditional navigation system, and a static virtual lead-vehicle (Pampel et al., 2019; Topliss et al., 2019). However, it is still unknown whether the virtual lead-vehicle performs better or worse compared to other world-fixed AR HUD concepts.

This study aimed to compare the world-fixed virtual lead-vehicle and a different world-fixed AR HUD concept with respect to their results on lateral driving performance and mental workload. Thirty-three participants participated in a within-subject driving simulator study, in which participants followed the navigation instructions of four different AR HUD concepts. The following AR HUD concepts were presented: a baseline concept that only showed screen-fixed speed and navigation information, two world-fixed virtual carpet concepts (one with longitudinal information and one without), and a world-fixed virtual lead-vehicle concept. The world-fixed concepts were all presented together with the screen-fixed speed and navigation information. It was expected that the virtual lead-vehicle would lead to a lower mental workload and lower lane position variability compared with the virtual carpet. Similarly, the virtual carpet with longitudinal information was expected to have a decreased mental workload and lane position variability compared with the virtual carpet without longitudinal information. In addition, exploratory measures were performed with respect to longitudinal driving performance (i.e., average speed, and speed limit exceedance), and subjective indications of safety, user experience and driving performance.

The results suggest that the virtual carpet is the most promising world-fixed AR HUD concept. This is due to the lower mental workload, higher user experience, higher perceived safety, and higher ranking for driving performance for the virtual carpet (basic) compared to the virtual lead-vehicle. However, for the objective driving performance measures between the virtual lead-vehicle and the virtual carpet no significant differences were found. The longitudinal information present in both the virtual lead-vehicle and the virtual carpet (longitudinal) caused a decrease in speed limit exceedance compared to the other two concepts, but there was no significant difference. Furthermore, mental workload increased with the addition of longitudinal information, but probably did not cause an overload since lateral driving behaviour was not affected. In addition, other measures did also not show a difference between the virtual carpet with and the virtual carpet without longitudinal information. However, since mental workload was slightly increased, it is suggested to investigate the effects on driving safety further before implementing longitudinal information in an AR HUD concept. Next to that, we also encourage researchers to investigate the virtual lead-vehicle concept further, as the virtual lead-vehicle might still be advantageous on complex intersections as was found by Topliss et al. (2019). Besides that, some of virtual lead-vehicle's behaviour in this study could be improved and should be explored further.

**Keywords:** augmented reality head-up display, car navigation systems, world-fixed, virtual lead-vehicle, virtual carpet, longitudinal information

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# 1. INTRODUCTION

Although automotive research topics are increasingly focusing on automated driving (Ayoub et al., 2019), it is expected that manually driven vehicles will be present on the road for decades (Calvert et al., 2017). Therefore, the design of in-vehicle devices such as navigation systems remains of high importance. The majority of in-car navigation systems are currently screens positioned in the center of the dashboard, i.e., between the passenger and the driver seat. This causes drivers to take their eyes off the road when they look for navigation information (Kun et al., 2009). A solution to this is now widely available, in which navigation information is presented on a Head-Up Display (HUD). A HUD is a transparent display or projection on the windshield that projects visual information into the primary field of view of the driver (Betancur et al., 2018). The visualizations differ per HUD concept, but they often include turn-by-turn navigation information, speed limit indications, and the driver's current speed. Compared to the navigation displays positioned in the center of the dashboard, HUDs reduce the eyes-off-the-road time, which is desirable as it decreases the possibility of a traffic accident (Liu & Wen, 2004; Smith et al., 2017). However, the high proximity of HUD information and real-world road information also increases the risk of cognitive tunnelling, meaning that drivers would effortlessly shift their focus between the navigation information and the real-world information, as the information is presented very close to each other (Bossi et al., 1997; Hagen et al., 2005).

A novel approach towards in-car navigation systems is the use of Augmented Reality (AR), in which virtual navigation information is projected on the car's windshield. AR navigation solutions have the opportunity to map information directly onto the relevant real-world situation, making the abstract information easier to interpret (Charissis & Papanastasiou, 2010; Park et al., 2013; Pauzie, 2015). In the car industry, design ideas for AR navigation systems are increasing. The AR HUD design of Panasonic Automotive is a good illustration of the state-of-the-art vision on these navigation systems (Panasonic Automotive, 2021) (Figure 1). It includes both screen-fixed (i.e., the bottom part of the virtual display containing navigation and traffic information) and world-fixed information (e.g., the route line and the object recognition features, all mapped onto the relevant road information). However, scientific studies about the designs of AR HUD navigation systems are lacking, which is a potential threat for the safety of users of these navigation systems when these systems become more common in vehicles (Betancur et al., 2018; Gabbard et al., 2014).

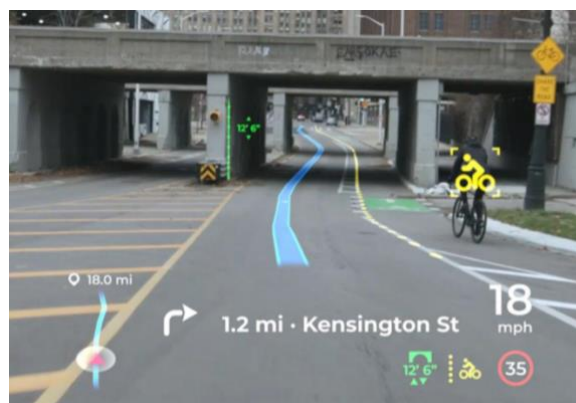


Figure 1. AR HUD design of Panasonic Automotive, released in January 2021.

Several studies introduced and evaluated different AR HUD navigation concepts (e.g., Kim & Dey, 2009; Merenda et al., 2018). These concepts can be distinguished as a virtual carpet, a virtual cable, virtual arrows pointing to the next turn or other landmarks, a virtual lead-vehicle, and a superimposed map. This paper examines the effectiveness of using a virtual lead-vehicle as a world-fixed AR HUD navigation concept. The idea of following a virtual lead-vehicle as a means of navigation was first introduced in 2006 and arises from the assumption that following a friend's car to get to a desired unknown destination is a familiar principle to many people, and therefore could be a very natural navigation concept (McNabb et al., 2017; Narzt et al., 2006). According to Pampel et al. (2019)

following a virtual lead-vehicle can be beneficial above other forms of AR HUD information, as it includes less abstract information. A pedestrian navigation concept similar to the virtual lead-vehicle is also presented by Google Maps, in which pedestrians can follow a virtual fox that is showing the route (Stein, 2019).

However, since the introduction of the virtual lead-vehicle, it took 15 years before it was published in a paper. Since 2019, two studies have examined the effect of following a virtual lead-vehicle (Pampel et al., 2019; Topliss et al., 2019). In a recent video-based study, the virtual lead-vehicle was the most suitable shortly before, during, and immediately after a maneuver in the route (Topliss et al., 2019). With these results, Topliss et al. (2019) also performed a driving simulator study, in which participants had to take the correct exit at a complex junction (roundabout). This study compared two versions of an AR HUD virtual lead-vehicle concept and found that a dynamic virtual lead-vehicle would be preferred, as the static virtual lead-vehicle (i.e., a virtual lead-vehicle that “waits” at correct junction exit) led to lower confidence and higher subjective workload. In addition, compared to a screen-fixed arrow AR HUD concept, both virtual lead-vehicle concepts attracted more visual attention. Finally, the study by Pampel et al. (2019) compared a world-fixed lead-vehicle with a screen-fixed arrow and a traditional head-down navigation system. They found that navigation accuracy and confidence were higher for the world-fixed lead-vehicle compared to the screen-fixed arrow, while the screen-fixed arrow induced faster responses. Additionally, both studies found that distance-to-turn information would be appreciated alongside world-fixed AR HUD navigation systems, such that participants can create a global prediction of the route (Topliss et al., 2019; Pampel et al., 2019).

Although these studies show that the virtual lead-vehicle can be a proper navigation system, it is necessary to compare the virtual lead-vehicle with other forms of world-fixed AR HUD navigation concepts. In such a manner, we can be more confident that certain (dis)advantages of the virtual lead-vehicle are only due to the virtual lead-vehicle concept, and not merely because it was a world-fixed AR HUD concept. Therefore, this study investigates whether the virtual lead-vehicle concept has better, or worse lateral driving performance and lower mental workload compared to a different world-fixed AR HUD navigation concept. The virtual carpet seems to be the logical world-fixed AR HUD concept to compare with, as it is currently presented most often in the automotive industry. Additionally, the virtual lead-vehicle studies had other shortcomings, such as that the participants were not in control of the vehicle in two out of the three studies, which could decrease generalizability. Next to that, this study uses a driving simulator in which other vehicles are present, to improve the resemblance to a real-world situation compared to the study by Topliss et al. (2019). Furthermore, all world-fixed AR HUD concepts will be accompanied with screen-fixed navigation and speed information, such that participants are better able to create a global prediction of the route. Everything combined, this study aims to answer the following research question:

*RQ: What is the effect of different world-fixed AR HUD navigation concepts on lateral driving performance and mental workload?*

We assume that following a lead-vehicle is more common to people compared to following a carpet. Therefore, we expect that following the lead-vehicle requires less cognitive demands, thus leading to a decrease in mental workload compared to following a virtual carpet. In addition, the virtual lead-vehicle proposed in our study provides longitudinal feedback, i.e., driving towards the virtual lead-vehicle indicates speeding and driving away from the virtual lead-vehicle indicates driving too slowly. This is expected to decrease mental workload as well, since the driver no longer has to calculate the difference between the speed and speed limit for him/herself. For similar reasons, lane position variability, which is an indicator for lateral driving performance, is expected to be lower when following a virtual lead-vehicle compared to following a virtual carpet. As following a lead-vehicle is assumed to be more common, it is expected that drivers would be able to pay more attention to the driving task, causing less lane position variability compared to the virtual carpet. Thus, when comparing the virtual lead-vehicle with the virtual carpet, the following hypotheses were formulated:

*H1: The virtual lead-vehicle leads to a lower lane position variability than the virtual carpet.*

*H2: The virtual lead-vehicle leads to a lower mental workload score than the virtual carpet.*

As mentioned, an additional feature of the virtual lead-vehicle, but not of the virtual carpet, is that it provides longitudinal feedback. To test solely the effect of adding longitudinal feedback to a world-fixed AR HUD concept, the virtual carpet will be presented using two concepts: one with longitudinal information and one without. In this way, we can know whether a difference between the virtual carpet and virtual lead-vehicle might be caused by the presence of longitudinal feedback or something else. The following hypotheses were formulated to test these expectations:

*H3: The addition of longitudinal information to the navigation information leads to a lower lane position variability.*

*H4: The addition of longitudinal information to the navigation information leads to a decreased mental workload score.*

The hypotheses were tested by means of a driving simulator study in which participants experienced four different AR HUD navigation concepts, consisting of a baseline screen-fixed HUD condition, two world-fixed virtual carpet conditions, and a virtual lead-vehicle condition. The screen-fixed HUD present in the baseline concept was also present in the other three concepts and provided speed and turn-by-turn navigation information. The two virtual carpet conditions are distinguishable by the supplementary longitudinal information, i.e., one of the carpets changes its colour when the driver is speeding while the other carpet's colour remains blue. This makes it possible to evaluate whether the difference between the virtual carpet and virtual lead-vehicle is merely due to the longitudinal information of the virtual lead-vehicle. In the study, both longitudinal and lateral driving performance measures and subjective measures are performed to compare the different world-fixed AR HUD concepts, with lane position variability and mental workload as the primary measures.

## 2. LITERATURE AND AR HUD CONCEPTS

Since 1972, there is a positive trend in traffic deaths in the Netherlands. Traffic deaths decreased in 40 years from roughly 3000 deaths per year to approximately 650 deaths a year (Slachtofferwijzer.nl, 2020). This is likely due to several safety measures, such as passive-safety measures (e.g., seatbelt and airbags), active-safety measures (e.g., anti-lock brakes), and policy changes (e.g., requiring child restraints) (Kun, 2018). However, since 2014, traffic deaths have increased again, with 68% of its causes being distraction possibly caused by using mobile phones and other devices while driving (Hwangbo et al., 2016). Therefore, when exploring user interfaces both for primary and secondary driving tasks, it is of major importance to assess their driving safety and user experience (Kun, 2018).

This section aims to give a thorough understanding of the information needed create safe AR HUD navigation systems. First, an overview is provided of the primary and secondary driving tasks and which safety issues arise concerning these tasks. Next, the navigation task will be explained in more detail, after which the different navigation systems are covered, including their safety risks and benefits. Then, an overview of AR HUD design concepts and their related human factors, technological and safety concerns is provided.

### 2.1 Driving Tasks

First, the various driving tasks are identified which are the basis of driving performance measures. In manual driving, the driver is responsible for controlling the longitudinal and lateral position of the vehicle, monitoring the environment, responding to objects and events, planning maneuvers, and enhancing conspicuity (Harvey & Stanton, 2016). The driving task can also be explained through Michon's 3-level hierarchy of driving, which distinguishes between strategic, maneuvering, and control levels (Michon, 1985). In the first level, the strategic level, the driver generates plans and focuses on the desired trip destination. The next level is maneuvering, which focuses on more detailed decision-making to reach the strategic goals, such as speed selection or whether the driver should take a turn at an intersection. The third level, the control level, is concerned with the lateral task of maintaining lane position, and the longitudinal task of maintaining acceptable speed.

Besides properly executing the primary driving task, driving nowadays also includes several secondary tasks, such as following navigation instructions, setting up entertainment (e.g., music), or communication (e.g., calling or talking with a friend) (Harvey & Stanton, 2016). These secondary tasks are often presented on so-called In-Vehicle Information Systems (IVIS), which are menu-based systems that include many secondary functions, accessible via a single screen-based interface. Having all information in one display reduces the number of buttons on different locations on the dashboard. According to Kun et al. (2009), this is beneficial, as any visual output to the driver may constitute a potentially dangerous source of distraction. However, not all buttons should cease to exist, since simple control operations require simple usability instead of complicated touch screen paths using multiple steps to get the same result (Bach et al., 2008). Together with Advanced Driver Assistance Systems (ADAS) such as adaptive cruise control and lane centering, these systems aim to improve driver safety, performance, efficiency, and comfort (Pauzie, 2015). The use of these in-vehicle systems has increased significantly, with four times more in-vehicle devices in 2007 than in 1954 (Kern et al., 2010).

To investigate objective driving performance, both longitudinal and lateral measures can be performed using driving simulators, test tracks or on-road experiments (Kun, 2018). In general, worse performance on any of the measures reduces driving safety. The paper of Kun (2018) provides an overview of the measures: the number of collisions, speed, lane position variability, crossing of lane markings, reaction-time measures, and physiological measures such as pupil diameter or the measurement of electro-dermal activity (Kun, 2018). In addition, many subjective measures have also been used to investigate the use of in-vehicle systems. Examples of subjective measures are mental workload measured (e.g., NASA-TLX (NASA Task Load Index) or DALI (Driving Activity Load

Index), (Hart & Staveland, 1988; Pauzie, 2008)), user experience (e.g., AtrakDiff, UEQ (User Experience Questionnaire and meCUE (Díaz-Oreiro et al., 2019; Schrepp et al., 2014))), trust (e.g., Lee & See, 2004) or acceptance (e.g., TAM (Technology Acceptance Model), (e.g., Lee et al., 2003)), self-constructed Likert-type questions measuring perceived safety or other subjective questions (e.g. Cho et al., 2017), or self-constructed ranking questions (e.g., Pärsch et al., 2019)). The measures used in this study are explained in more detail in Section 3.5.

## **2.2 Human Factors Issues: Maintaining Safety and Usability**

Although in-vehicle systems aim to improve the driving experience and are meant to enhance mobility, it has also been shown that they distract the driver from the primary driving task, which could decrease driving safety (Jensen et al., 2010). Both for primary and secondary driving tasks, drivers use multiple mental resources (e.g., physical, visual or auditory) (Wickens, 2002), with each of the channels having their own threshold for a maximum capacity. If the combination of resources needed for both the primary and secondary tasks together exceeds the available resources, mental workload will be too high, causing the driver to focus too much on secondary tasks, decreasing the perception of information needed to safely control the vehicle. For the driving task specifically, the perceptive visual channel is the most essential, as 90% of the information needed to perform the driving task flawlessly is perceived by the visual channel (Castro, 2008). When designing user interfaces of in-vehicle systems, it is therefore important to investigate the amount of mental resources needed to use them, such that they will not negatively affect driving performance (Kun, 2018). It may be the case that when poorly designing a navigation system, it can just be as distracting as using paper maps (Dingus & Hulse, 1993).

Next to minimizing mental workload, other human factors or ergonomics principles should be taken into account when designing usable and safe in-vehicle systems. In general, these principles are based on scientific studies and used to design object, systems, and environments for human use (Caple, 2010). In the automotive context, often discussed principles are eyes-off-the-road time, legibility, cognitive tunneling, cognitive distance and occlusion of relevant information. These principles will be discussed below when describing the different navigation systems.

## **2.3 In-Car Navigation**

Since the first Global Positioning System (GPS) was introduced in 1988, navigation systems have been presented in many different shapes and concepts (Azuma et al., 1994). Where navigation systems first grew significantly as portable devices (Figure 2.a), the current GPS market is mainly divided into smartphone applications (Figure 2.d) and in-dash in-vehicle systems (Figure 2.b). These systems all have their strengths and weaknesses related to usability and driving safety, which will be covered in this section.

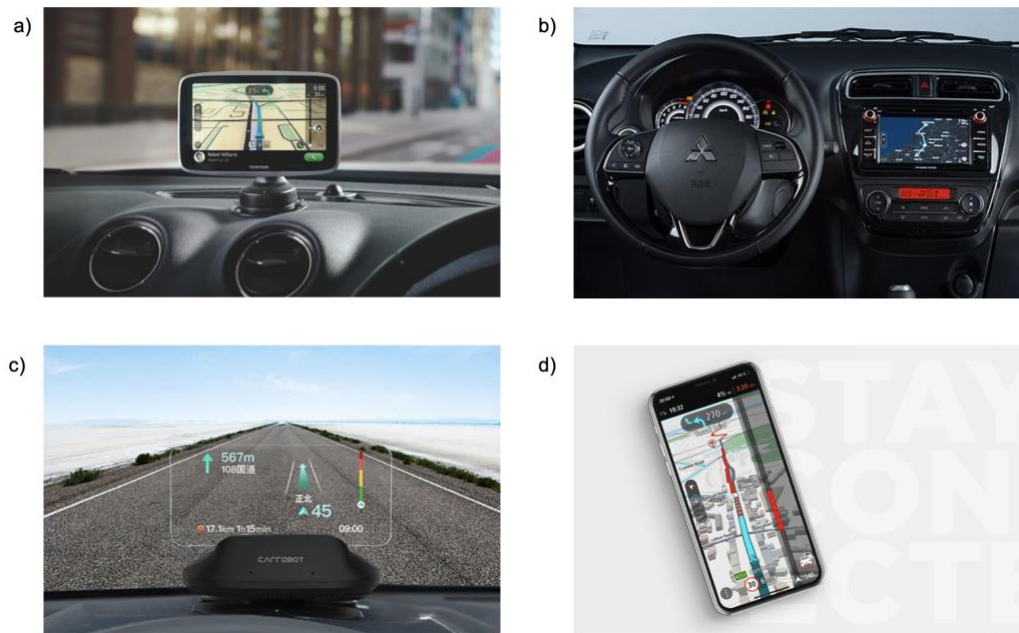


Figure 2. Different navigation system concepts. a) Traditional navigation system (TomTom, n.d.); b) navigation system positioned in the center of the dashboard (TomTom, n.d.); c) Head-Up Display (Xiaomi, n.d.); d) Smartphone application (TomTom, n.d.)

### 2.3.1 The Navigation Task

Burnett (1998) extended Michon's 3-level driving task model (Section 2.1) by integrating the driver's goals and requirements in a navigation task (Burnett, 1998; Michon, 1985; Ross & Burnett, 2001). This model includes six overlapping stages: trip planning, preview, identify, confirm, trust and orientation (Figure 3). First of all, in the trip planning stage, the driver specifies the intended destination(s) and plans a route. In the preview stage, the remaining time and distance until the next maneuver is summarized into a mental picture, together with preparatory knowledge about the required position on the road. In the identify stage, the driver identifies the location of the next maneuver, how to arrive at this location, while using the correct speed and positioning at the final approach. In the confirm stage, the driver establishes whether the maneuver was taken correctly. Finally, both the orientation and trust stages take place throughout the navigation task but are mainly present at the end of the task. Here, in the trust stage, the driver gains reassurance that the correct route is being driven, and in the orientation stage spatial awareness is gained in relation to general surroundings and final destination (Burnett, 1998; Ross & Burnett, 2001).

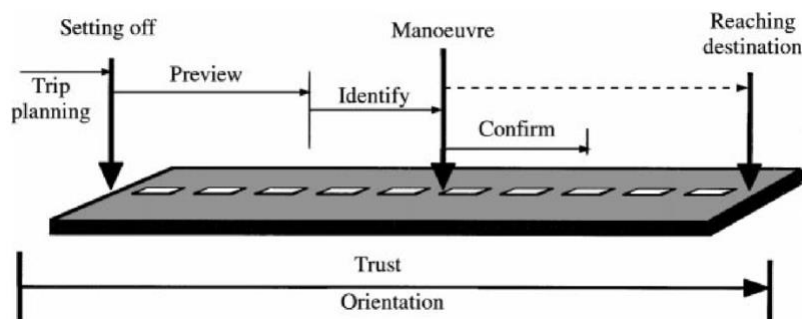


Figure 3. The six stages of the navigation task (Ross & Burnett, 2001)

### 2.3.2 Early Way-Finding Methods

Before the existence of electronic navigation devices, multiple methods were used to navigate when travelling on unknown roads, e.g., road signs, maps, pre-written lists of instructions, the passenger

seat navigator, and following a friend's car (Burnett, 1998; McNabb et al., 2017). However, these methods have shown to have many deficiencies. For example, reading a map while driving is associated with a high task workload and large percentages of time spent with the eyes off the road (Wierwille et al., 1989). Additionally, written notes are often misread by the driver and are very impractical once a navigational mistake is made (Wickens, 2002). Therefore, for both monetary and efficiency reasons for the driver and countries at large, it was of high interest to find a more efficient way-finding method (Burnett, 1998).

### **2.3.3 In-dash Navigation Systems and Portable Navigation Systems**

Nowadays, in-car GPS based navigation systems, in the form of portable units or in-car dashboard navigation systems, are a common part of driving (Brown & Laurier, 2012). These systems display maps and provide turn-by-turn navigation information, using visual and audio guidance. By means of a positioning system and a big database of maps, the suitable route for the driver is calculated and displayed. Additionally, spoken directions are provided to decrease the number of times needed to glance at the visual display and to decrease the visual mental workload (Kun et al., 2009). As mentioned, the first navigation systems were separate devices that could be attached to the dashboard or onto the windshield, such as the TomTom GO (Figure 2.a) (TomTom, n.d.). These systems were used extensively, but they often had a limited screen size and failings with respect to their sensors and map information (Brown & Laurier, 2012). Besides that, it was even found that an audio-only navigation condition resulted in higher driving performance compared to using a GPS navigation system that was placed at the lower center of the windshield, showing a potential need for improvement of the graphical display (Jensen et al., 2010).

With the rise of smartphones, people gained access to GPS technology and high-resolution displays in the hand-held mobile phones, which quickly led to cheap and high-quality navigation software (Figure 2.d) (Jokinen, 2008). Next to that, car manufacturers started creating in-vehicle systems, which included in-dash navigation systems (Figure 2.b). As explained before, an advantage of the in-vehicle systems such as in-dash navigation system, is that all secondary task information is presented in the same location, reducing the information clutter when many displays would be present. Presenting information in the same location is beneficial, as any visual output to the driver may be a potentially dangerous source of distraction (Kun et al., 2009). However, this does not mean that all buttons implemented in the touchscreens, since some simple control operations require that they are quickly reachable which is more difficult with touchscreen interfaces. A remaining disadvantage of in-dash navigation systems is that they are located far away from the road situation, and often require extensive processing, causing drivers to take their eyes off the road situation (Dingus et al., 1988). Another problem of navigation systems is that the distance between the physical space (e.g., the real road environment) and virtual information space (e.g., the virtual navigation information) is quite large (Kim & Dey, 2016). Because of this large distance, time and cognitive effort are required to adjust attention from one space to another (Kim & Dey, 2009; Pauzie, 2015). An additional critique towards navigation devices, and a problem of automation in general, is the loss of engagement with the environment and others caused by an increased immersion with the virtual environment (Leshed et al., 2008). This decreased immersion with the real-world might also cause drivers to lose way-finding skills and become too dependent on the technology (Brown & Laurier, 2012). Therefore, navigation solutions that are enhancing the immersion with the real environment should be supported.

### **2.3.4 Head-Up Display**

Although still a novel technology for many drivers, Head-Up Displays (HUDs) are currently being produced both as separate device and as a built-in car feature. HUDs are transparent displays or projections on the windshield which project visual information into the primary field of view of the driver (Figure 2.c) (Betancur et al., 2018). This information often shows the speed of the car, the speed limit on the road, navigation information, and ADAS information such as settings for cruise control or lane departure warnings (Citraen, 2021; Mazda Motor Corporation, 2021; Volkswagen AG, 2021).

HUDs have shown to be valuable when keeping eyes on the road, and to support faster responses to highway events and emergency-related information (Liu & Wen, 2004; Smith et al., 2017; Srinivasan & Jovanis, 1997). Besides that, speed control is also more consistent with a HUD than a conventional Head-Down Display (HDD) (Charissis & Papanastasiou, 2010; Liu & Wen, 2004). Additionally, less navigation errors and speeding offenses are made when using a HUD that provides turn-by-turn navigation information and information about speed limit exceedance, compared to using a HDD (Jose et al., 2016).

However, similar to the in-dash navigation systems, HUDs also face the problem of cognitive distance. Although the information is presented close to the driver's field of view, drivers still need to switch between the virtual information space and the physical space, which results in difficulties with vehicle control actions and decision-making (Pauzie, 2015). Another challenge of HUD navigation systems is cognitive tunneling, or also called cognitive capture, in which people focus on a task that has a low priority (e.g., reading the navigation system information) at the expense of focusing on a task with higher priority (e.g., driving) (Pauzie, 2015). As HUD systems have been adopted in aircrafts already for a while, several studies have investigated the effect of cognitive tunneling when using HUD systems. It was found that pilots using a HDD were more likely to observe an aircraft taxiing on a runway than pilots using a HUD (Fischer et al., 1980). Additionally, pilot's response time to unexpected events decreased when using a HUD (Wickens & Long, 1995). Similar for cars, HUDs can cause cognitive capture, especially those using textual output (Bossi et al., 1997; Hagen et al., 2005).

## **2.4 Augmented Reality Head-Up Display**

A solution to cognitive tunneling and cognitive distance problem is to spatially map navigation information to the object of reference, which is known as AR HUD (Pauzie, 2015; Ververs & Wickens, 2000). This introduces new opportunities for fast and efficient information presentation, as it reduces the number of glances necessary to obtain information relevant for the driver. Additionally, it might improve the engagement with the virtual environment, which had decreased with the emergence of traditional navigation systems (Brown & Laurier, 2012; Leshed et al., 2008). This section will start by introducing the different AR HUD navigation design concepts, both in terms of navigation system and other opportunities, followed by the corresponding human factors challenges. In addition, this section will discuss the different technological shapes in which AR HUDs are currently being presented, after which a few of AR HUD's technological challenges will be covered as well.

### **2.4.1 AR HUD Navigation Design Concepts**

A distinction can be made between screen-fixed and world-fixed AR HUD concepts. Screen-fixed AR HUD concepts resemble the conventional HUD concepts, meaning that the visual information is rendered at a fixed location on the display screen (Gabbard et al., 2014). Conversely, world-fixed AR HUD concepts are attached to specific objects in the driver's field of view, which is specifically useful when adding navigation instructions, driving instructions and hazard warnings. World-fixed AR concepts also have the opportunity to reveal hidden exits or roundabouts (Narzt et al., 2006). The screen-fixed concept is specifically useful for information that is not directly attached to objects in the real-world, such as distance-to-turn information, vehicle speed, fuel level and engine temperature (Figure 4.a), or can be used to present an arrow when approaching a maneuver point (Figure 4.b) (Pärsch et al., 2019; Pfannmüller et al., 2015). Compared to several world-fixed concepts, Pärsch et al. (2019) found a screen-fixed concept to be ranked best, which they explained by the fact that users often react negatively to novel devices (Dillon & Morris, 1996). On the contrary, Pfannmüller et al. (2015) found that for both 5-point Likert scale items distraction and annoyance, a screen-fixed HUD was rated significantly worse compared to a world-fixed arrow and boomerang concept.

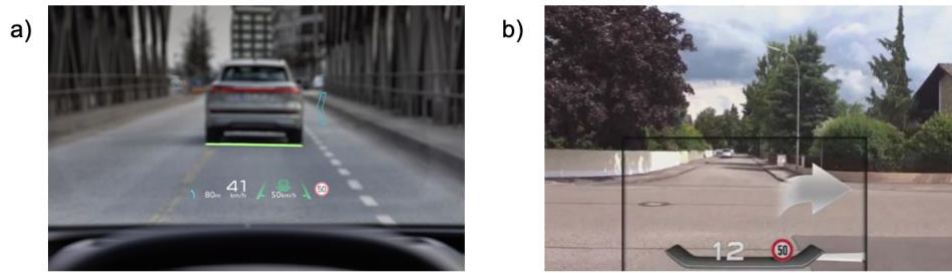


Figure 4. Overview of screen-fixed AR HUD navigation concepts. a) navigation and other driving information such as speed and lane departure warnings (Audi, n.d.); b) arrow that appears when approaching a maneuver point (Pfanmüller et al., 2015).

It was found that, generally, world-fixed AR navigation effectively directs attention to real-world targets, however, it is difficult to detect them at a distance (Merenda et al., 2018). On the contrary, screen-fixed navigation is more salient as it is always present at the same location on the display, but it is more difficult to interpret the distance to a specific target, such as determining which road to take for the upcoming turn. Specifically for motion-based AR cues, Merenda et al. (2018) indicate that more work is needed to understand how to best design these cues.

The remainder of this section will give an overview of the world-fixed AR HUD concepts studied in literature.

#### A. Virtual Carpet

Currently, a world-fixed virtual route line seems to be the state-of-the-art when it comes to AR HUD navigation designs. The AR HUD design of Panasonic Automotive (Figure 1) also uses a virtual carpet to indicate what road to follow. A recent study has investigated five different AR HUD design concepts, including three virtual carpet designs, a boomerang concept, and a screen-fixed arrow (Pärsch et al., 2019). They found that more navigation mistakes were made with the carpet with maximum coverage (Figure 5.a) and with the lane markings concept (Figure 5.c). Additionally, it was found that the perceived disturbance from coverage was highest with the carpet with maximum coverage (Figure 5.a) and with the boomerang concept (Figure 5.d). As mentioned before, participants ranked the conventional HUD as the best navigation system (Figure 5.e), which they explained by the fact that users often react negatively to novel devices (Dillon & Morris, 1996).

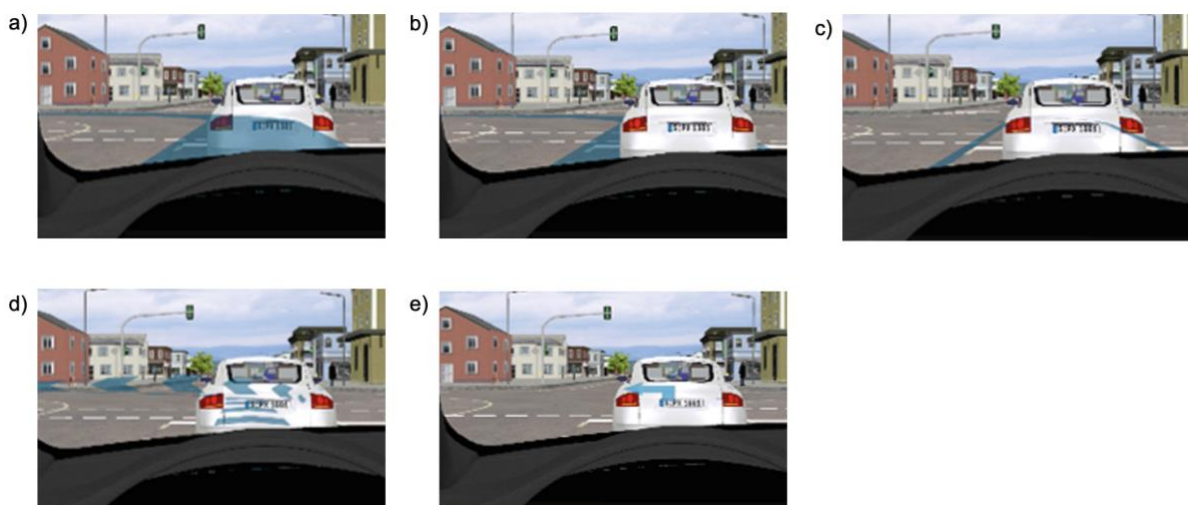


Figure 5. Overview of AR HUD navigation concepts. *a)* carpet with maximum coverage; *b)* carpet with minimal coverage; *c)* lane markings; *d)* boomerang; *e)* conventional arrow. (Pärsch et al., 2019)

The virtual carpet has also been presented as a boomerang or world-fixed arrow concept by Pfanmüller et al. (2015) (Figure 6). Here they found that when comparing a world-fixed concept similar to Figure 5.a with a screen-fixed arrow or with a slightly tilted world-fixed carpet, that the tilt is not recommended for AR HUD systems. This was based on answers to 5-point Likert scale items; attractiveness, accuracy, functionality, clearness/ambiguousness, annoyance, distraction, interpretability and intuitiveness. No significant difference, however, was found between the screen-fixed arrow and the world-fixed concepts. Only for clearness/ambiguousness, interpretability and intuitiveness, a significant preference for the boomerang concept was found over the arrow concept, which Pfanmüller et al. (2015) guess can be due to masking or attentional and cognitive capture.



Figure 6. Overview of AR HUD navigation concepts presented by Pfanmüller et al. (2015). *a)* world-fixed arrow with maximum coverage; *b)* tilted world-fixed boomerang; *c)* screen-fixed arrow.

### B. Virtual Cable

Similar to the virtual carpet, with the virtual cable concept, drivers also follow a route line (Figure 7). However, in case of the virtual cable, the route line is hovering above the vehicles present on the road, at a height of about 2 meters. In a study by Medenica et al. (2011), the virtual cable was preferred by the participants above an AR video concept and a traditional navigation system. Additionally, it resulted in the highest driving performance, and the highest time spent looking at the road.



Figure 7. Virtual cable concept (Medenica et al., 2011)

### C. Landmarks / arrows

Another implementation of AR HUD for navigation is the use of landmarks, which is consistent with basic human wayfinding strategies (Figure 8). In earlier research, the use of landmarks was mainly purposed for voice messages (Gary et al., 2020). However, nowadays its opportunity for AR HUD navigation is being investigated. For example, in a study by Bolton et al. (2015) comparing different AR HUD concepts, it was found that when distance-to-turn approaches are accompanied by landmark-based navigation, decision times are reduced and navigation success rates are enhanced. However, concerns are still present about its distraction caused by presenting additional information to drivers. Additionally, more research is desired into the impact on driving performance, both in the simulator and in the real-world.



Figure 8. Landmarks / arrows concept (Bolton et al., 2015)

#### D. Superimpose map on road information

To provide future navigation information, Kim and Dey (2009) proposed a concept in which they superimposed the navigation information on the driver's view of the actual road using a full windshield HUD (Figure 9). This concept provided drivers with local route guidance, but also with global awareness of the roads ahead, while keeping their eyes on the road. Their proposed concept was received very positively by both younger and older participants. Additionally, compared to a traditional navigation system, navigation performance and divided-attention related issues were better for the superimposed map concept compared to a traditional navigation system. However, qualitative responses indicated that the information was not always clear, such that some drivers were confused about whether they could continue to go straight, regardless of the state of the traffic lights.

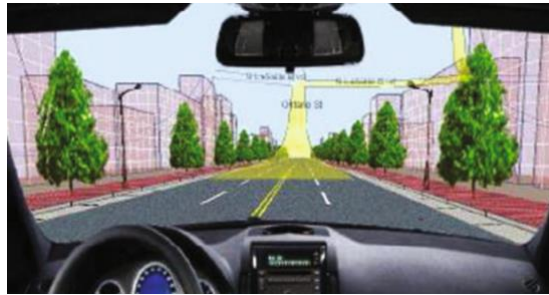


Figure 9. Superimposed map concept (Kim & Dey, 2009)

#### E. Virtual Lead-vehicle

As introduced earlier, the virtual lead-vehicle concept virtually presents a vehicle in the real-world, such that the driver can follow it to get to a desired unknown destination (Figure 10) (Narzt et al., 2006). According to Pampel et al. (2019) following a virtual lead-vehicle can be beneficial above other forms of AR HUD information, as it includes less abstract information (e.g., interpreting icons, distances, and directions). Since 2019, several studies have examined the effect of following a virtual lead-vehicle. In a recent video-based study, participants watched a video in which they followed a real-world vehicle for 15 minutes while using a steering-wheel, with or without additional auditory or visual navigation information present (Topliss et al., 2019). When entering unfamiliar traffic junctions, or when the road situation was very complicated, drivers preferred to follow the lead-vehicle instead of looking at the navigation instructions. In addition, 14 out of 24 participants indicated their preference for using the lead-vehicle as a primary aid when they were in the Identify and Confirm stages of the navigation task, even with a traditional navigation system present. However, participants also indicated that following the lead-vehicle alone lacks global information about future turns, making it difficult to create a global prediction of the route. This caused difficulties for the Preview and Orientation stage of the navigation task, as well as the Confidence stage. In addition, participants glanced significantly more often to the lead-vehicle compared to the visual navigation information, but this did not result in lower hazard detection. Altogether, the results of the first study of Topliss et al. (2019) show that a virtual lead-vehicle could be most suitable just before, during and immediately after a manoeuvre in the route.

Next to the video-based study, Topliss et al. (2019) performed a driving simulator study in which participants followed a virtual lead-vehicle projected on a HUD. The virtual lead-vehicle moved through the junctions ahead of the participant's vehicle like a normal real-world vehicle would. Supported by the results of their first study, in this second study participants had to take the correct exit at a complex junction (roundabout). It was found that navigation accuracy and driving performance did not differ between a dynamic virtual lead-vehicle, a static virtual lead-vehicle (i.e., a virtual lead-vehicle that "waits" at correct junction exit and does not move through the junction) and a screen-fixed arrow concept. Additionally, the static virtual lead-vehicle resulted in lower confidence and greater subjective workload, indicating a preference towards the dynamic virtual lead-vehicle. Objective eye-glance measures, however, indicate that in the dynamic virtual lead-vehicle condition the single glances towards the virtual lead-vehicle were longer and the number of glances towards the HUD was also highest. Potentially, this is a disadvantage of the dynamic virtual lead-vehicle, because less glances to the road environment might result in a decrease of situation awareness.

Another study including an AR virtual lead-vehicle was performed by Pampel et al. (2019), in which participants were shown simulated routes and had to indicate the correct turn using the cues of the different navigation systems. Next to a world-fixed virtual lead-vehicle that was positioned 50 ft in front of the driver, used its indicator lights before the next turn and stopped at the correct side road before disappearing, this study presented a screen-fixed arrow presented on an AR HUD and a traditional head-down navigation system. It was found that navigation accuracy and confidence was higher for the world-fixed lead-vehicle compared to the screen-fixed arrow, while the screen-fixed arrow induced faster responses. As seen in the study by Topliss et al. (2019), participants indicated that some distance-to-turn information would be appreciated alongside the HUD conditions.



Figure 10. Virtual lead-vehicle concept (Topliss et al., 2019)

#### 2.4.2 Alternative AR HUD Concepts

Besides using AR for navigation, there are many other applications for AR in vehicles (Riegler et al., 2020). For example, Points-of-Interest (POIs) such as parking spaces, public transport, restaurants, and gas stations could be highlighted using AR (Schouten, 2020). Additionally, feedback about the system state of autonomous cars in the form of AR route information significantly increases trust in the vehicle (Von Sawitzky et al., 2019). Next to that, Tönnis et al. (2007) found that displaying the drive path in AR can improve lane-keeping, which is especially desirable in bad weather conditions and darkness (Tönnis et al., 2007). AR can also be used for lane changing tasks. It was found that using lane change AR increased the number of times that the drivers used the brake pedal to reduce speed (Lorenz et al., 2014). However, drivers were also more often first attracted to the AR information, than that they were looking in their side mirror to confirm their lane change. Another implementation of AR HUDs is the detection of critical road events. Displaying critical road events on the windshield can help hazard detection with an advantage of 100-ms compared to using a HDD (Weihrauch et al., 1989).

AR can also be used as a research platform, such as ARCAR developed by Ghiurău et al. (2020), in which drivers are in full control of a moving vehicle, while experiencing a mixed reality environment by wearing a headset. This mixed reality experience provides the opportunity to investigate

topics such as driver's attention and reaction to different behaviors of the car when a virtual obstacle is injected onto the road. Another example of research performed with AR is the study by Smith et al. (2020), in which participants were following a lead car that unexpectedly braked while completing secondary tasks on AR HUDs or HDDs.

#### **2.4.3 Human Factors Challenges with AR HUD Navigation Concepts**

As mentioned above, human factors or ergonomics principles should be taken into account when designing usable and safe in-vehicle systems such as an AR HUD navigation system. It is important that the driver is aware of his or her surroundings, such that the driver can react to (hazardous) situations on the road. Situation awareness consists of three stages: perception, comprehension, and projection of their future status (Endsley, 1988). However, for AR HUDs there might be a chance that the driver focuses too much on the virtual information, and therefore does not perceive important information from the road environment. More specifically, when observers are focused on some other object or event, they often fail to notice salient and distinctive objects (Simons, 2000). This inattentional blindness can be a threat for the safety of AR HUD navigation systems. Inattentional blindness is something drivers should be made aware of, such that they keep focusing on the road instead of the AR HUD display. Additionally, displays should be designed such that they minimize clutter, i.e., displays should reduce information that intervenes or confuses other information or that is not relevant for the primary driving task (Pankok & Kaber, 2018). Features that decrease the saliency of a target relative to a background or color have a negative impact on human visual search (Pankok & Kaber, 2018). The size of the display region, the size of the target of interest, the number of objects, the density of objects, the complexity of objects or similarity of objects also affect display clutter (Kaber et al., 2008). For example, the study by Currano et al. (2021) shows that situation awareness decreases with increased HUD complexity when driving in autonomous vehicles. As a result of increased clutter, response time to a task such as a potential road hazard increases as well. Specifically, when designing world-fixed AR HUD navigation systems, where the display information is presented on top of the relevant road information, it is relevant to analyze whether the benefits of providing information outweigh potential decreasing visual performance.

A challenge arising from visual clutter comes with the location and size of the HUD display. Compared to the scenery in aviation, in which navigation information is often presented in the primary visual area, the background of a HUD in a driving context is much more complex (Ward & Parkes, 1994). When presenting information on the car windshield in the primary visual area, there is a risk of masking any relevant road information that is needed to respond to unexpected traffic events. However, when simple information, such as a virtual speedometer is presented slightly downwards on the windshield, these masking effects don't seem to be a problem (Horrey & Wickens, 2004; Kiefer & Gellatly, 1996). Additionally, it was found that the preferred location for HUD information was not in the driver's center line of sight, but slightly left or right from it (Tretten et al., 2011). This however, is in contrast with the theory of cognitive tunneling, as locating information away from the primary visual area might result in focusing too much on the navigation information, instead of on the road information (Ward & Parkes, 1994). Therefore, research is now looking into AR HUD designs that are presented close to, or on, the primary visual area, but that are mapped in such a way that masking effects are minimal, or not present.

Another challenge arising from visual clutter is the design of AR HUD graphics. The study by Merenda et al. (2016) has investigated legibility of AR HUD designs against three different backgrounds (i.e. brick, pavement, and grass) and found the colors blue, green and yellow to perform more consistently and reliably on compared to brown, orange, pink, purple, and red. Challenges arise, however, when it comes to a changing background and different lighting and weather conditions (Gabbard et al., 2014). This interaction of backgrounds can affect the legibility of AR HUD graphics. A solution for this would be to adjust the color of the AR HUD conform the background change. However, colors are often used for specific features (e.g., lane change assist), which would make it very confusing when colors are changing either because of background change, or because of a built-in feature. Blue, however, seems to be a color that is not often used for specific in-built ADAS features,

and is also best recognized in different lighting conditions, leaving it as the best color for AR HUD design (Grünler, 2018; Merenda et al., 2016).

Finally, as indicated by Kim and Dey (2009), another need for AR HUD navigation systems is to provide the driver with future knowledge about the route ahead, similar to the information provided by traditional navigation systems. This can be provided by presenting distance-to-turn information by means of a screen-fixed AR HUD. Alternatively, a more extensive solution would be to use a superimposed map concept as presented by Kim and Dey (2009).

#### 2.4.4 Technological AR HUD Concepts

AR concepts do not only appear projected on the windshield, but they can also be an overlay over a video of the real-world driving situation, called AR video. However, in a driving simulator study by Medenica et al. (2011) it was found that participants preferred using the AR HUD compared to AR video and a traditional navigation system. Additionally, the AR video concept was associated with lower visual attention and worse driving performance. Similar findings were presented in a study by Haladijan (2020), in which participants were presented both an AR HUD and an AR video concept, and indicated that they were afraid that the AR video concept contained such detailed information that it would distract the driver from looking at the actual road. Another alternative for AR HUDs is AR glasses. AR glasses present the graphical information on top of the world via a display inside the wearable eyewear (Sawh, 2021). Research has looked into using AR glasses for video calling while driving a vehicle, arguing that one source of distraction coming from a phone call is the lack of shared context (Kun et al., 2019). AR glasses could overlay the video content rather close to the road information, which Kun et al. (2019) found to be not distracting. However, they recommend that more research should be performed before using AR glasses for the purpose of video calling.

#### 2.4.5 Technological Challenges of AR HUDs

The opportunities of AR HUD interfaces are currently still limited by technology possibilities. It is beyond the scope of this paper to provide a thorough overview, but a few of these challenges require an introduction (see Gabbard et al. (2014) for an overview). Currently, AR HUD interfaces still have a limited field of view, meaning that the navigation information cannot be presented on all locations on the windshield. This causes difficulty when presenting world-fixed AR navigation information, as some information might need more space to be able to overlay the real world. This was, for example, the case in a study performed by Wesselink & Schouten (2020), where it was found that an upcoming turn is sometimes invisible for the driver, as it is located outside the boundaries of the display. In a different study by TomTom B.V., they also found arrows being cut off by the boundaries of the display in such a way that it remained unclear which direction to go to (Haladijan, 2020). In a study by Pfanmüller et al. (2015), however, they investigated whether the design of the world-fixed arrow concept should be limited by the interface size. They compared an arrow concept that is cut off by the AR HUD boundaries, and one that is not (Figure 11). Interestingly, they did not find a significant difference between these conditions on 5-point Likert scale items; attractiveness, accuracy, functionality, clearness/ambiguousness, annoyance, distraction, interpretability, and intuitiveness.

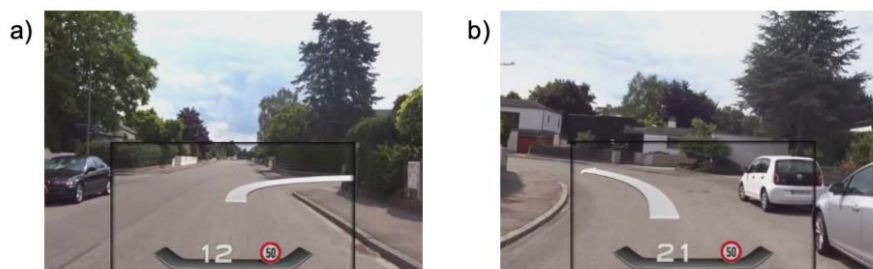


Figure 11. World-fixed AR HUD concepts as presented by Pfanmüller et al. (2015). a) cut-off; b) no cut-off.

Besides that, the cognitive distance can increase when the AR graphics are displayed at a different focal depth than the real-world objects they are referring to, causing drivers to accommodate at two different distances (Gabbard et al., 2014). Although the accommodation time needed for this type of AR HUD is less compared to other types of navigation systems, it is still a significant safety concern for driving tasks. Another challenge concerns the tracking of the vehicle, the relative position of the driver's head/eyes, and objects on the road. According to Gabbard et al. (2014), it can be quite certain that even with accurate tracking technology, there will be some noise in the system, for example caused by vibrations from road surfaces. It is not known yet how much noise is tolerable until world-fixed AR graphics are misunderstood and mistakenly referred to a different real-world object.

### 3. METHOD

#### 3.1 Design

The current experiment features a within-subject driving simulator study, covering four conditions with each a different design for an AR HUD navigation system. The four conditions are presented in Figure 12. The screen-fixed HUD information is present in all four conditions, whereas additional world-fixed navigation is present in all conditions except the baseline (condition 0) (Figure 12.a). A virtual carpet, functioning as a route line, is presented in both condition 1 and condition 2, for which the carpet in condition 2 changes color when the driver is exceeding the speed limit (Figure 12.b and Figure 12.c). In condition 3, the driver follows a virtual lead-vehicle as a means of navigation (Figure 12.d). The color change of the virtual carpet (longitudinal) is added to investigate whether the addition of longitudinal information influences the difference between the virtual carpet and the virtual lead-vehicle. All four condition designs are explained in greater detail in Section 3.4.2. A latin-square design was used to avoid order effects (Voss et al., 2017).

For each trial, both subjective and objective measures were performed. The subjective measures were obtained by a questionnaire, whereas the objective measures were obtained by logging data from the driving simulator while the participant was driving. For each participant, the experiment lasted about 60 minutes.

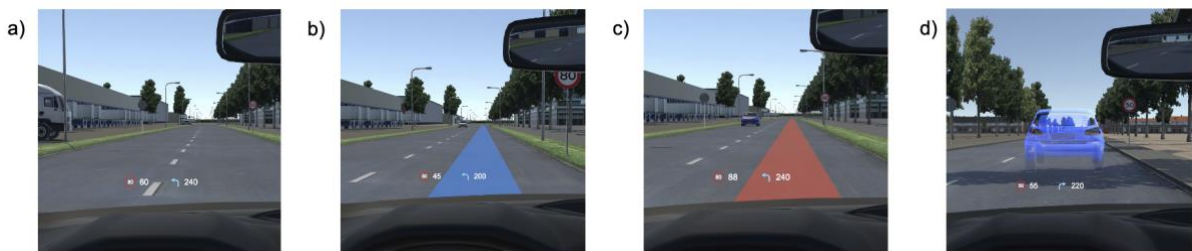


Figure 12. AR HUD concepts presented in this study: a) condition 0 - baseline concept; b) condition 1 - virtual carpet (basic) concept; c) condition 2 - virtual carpet (longitudinal) concept; d) condition 3 - virtual lead-vehicle concept

#### 3.2 Participants and Recruitment

The participant sample consisted of 35 participants. However, two participants experienced motion sickness after the first or second trial, which is why the data for these participants is not included. The remaining participant consisted of 15 male and 18 female participants, which were aged between 19 and 59 ( $M = 24.8$  years,  $SD = 6.6$  years). All participants were members of the HTI database of the Eindhoven University of Technology, and were paid €10,- (or €12,- for people outside of the university), for their participation. They were all in the possession of a driver's license for 1 to 40 years ( $M = 6.5$  years,  $SD = 6.6$  years), which could originate from any country to be eligible for participation. Three of the participants indicated that they drove for most of their life on the left side of the road, and 32 participants on the right side. As recruitment criteria, participants were demanded to drive at least once every three months. Three participants indicated that they did not drive in the last three months prior to the experiment.

The sample size was based on power analysis: based on an expected effect size of 0.40 (Topliss et al., 2019) a power of 0.9 is achieved for an omnibus test at  $n = 35$ .

#### 3.3 Experiment Setup

The experiment was conducted using a driving simulator setup from Green Dino B.V. (Green Dino, n.d.), located at the Technical University in Eindhoven (Figure 13.a). Next to the driving

simulator, a table with a laptop was present containing the survey (Figure 13.b). While the participant was driving, the experimenter was able to see what was happening in the virtual driving simulator environment via a monitor (Figure 13.c).

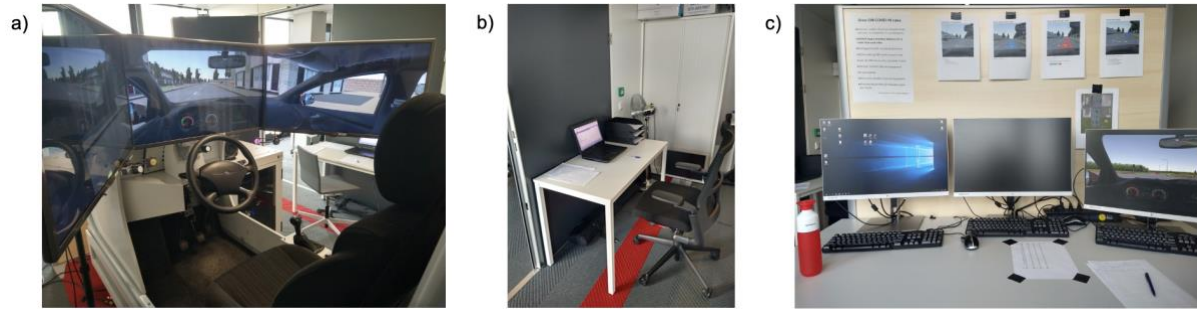


Figure 13. Experiment set-up. a) driving simulator; b) desk with laptop containing the survey; c) experimenter set-up

### 3.4 Driving Scenario

#### 3.4.1 Design of Environment

For the experiment, the Dutch city scene ‘CityPlusHighway’ from Green Dino B.V. was used. Following the taxonomy of Fastenmeier (1995), this environment can be classified as a rural environment. Compared to highway or urban environments, rural environments show the most individual differences with respect to driving measures such as crash occurrence and lane position variability (Michaels et al., 2017). The environment includes many different buildings, trees, traffic signs and details such as playgrounds, windmills, gas stations and non-moving pedestrians. Next to that, the environment included other traffic driving according to general driving rules. Figure 14.a shows an example of the scenery, and Figure 14.b shows the top-view of the environment including the navigation route. The route was presented to participants in both directions, resulting in route A and B, both taking approximately 7 minutes to drive. These routes both include eleven intersections including crossroads with and without traffic lights, T-junctions and a roundabout, and speed ranging from 50 to 80 km/h (Table 1). For route A, there were two maneuvers to the right, four to the left, one roundabout maneuver to the right, and four straight maneuvers. Similarly, route B consisted of four maneuvers to the right, two to the left, one roundabout maneuver to the left, and four straight maneuvers.

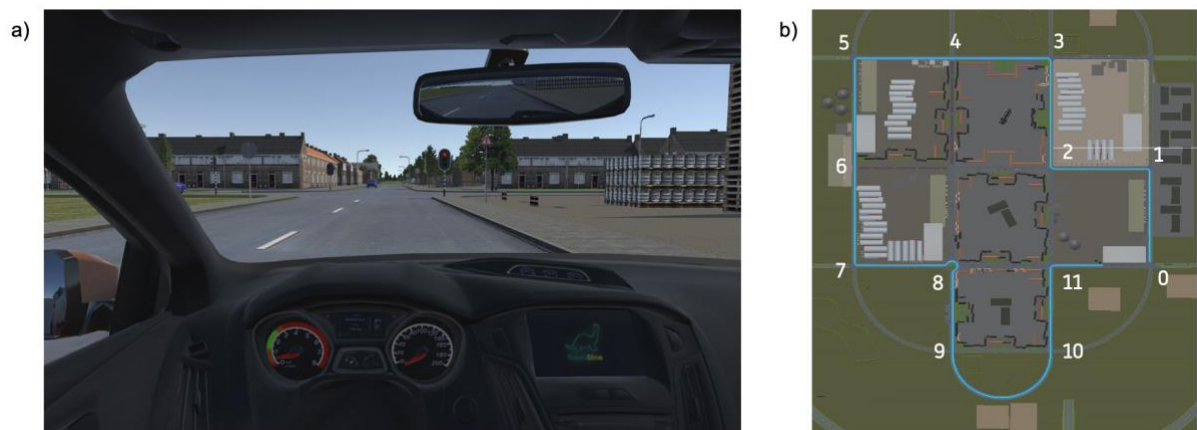


Figure 14. Virtual environment in Unity3D. a) interior of the car and example of environment; b) top-view of environment in which the blue line indicates route B. Numbers indicate the intersection number.

Table 1

*Route details*

| Junction type                       | Count |
|-------------------------------------|-------|
| Intersection                        | 11    |
| - Crossroad with traffic lights     | 5     |
| - Crossroad without traffic lights  | 3     |
| - T-junction without traffic lights | 2     |
| - Roundabout                        | 1     |
| Road                                | 12    |
| - Straight 50 km/h                  | 7     |
| - Straight 70 km/h                  | 1     |
| - Straight 80 km/h                  | 3     |
| - Curved 80 km/h                    | 1     |

**3.4.2 Design of AR HUD Navigation Concepts**

The four different AR HUD navigation concepts are depicted in Figure 12. The design of these four conditions can be distinguished in a screen-fixed design and multiple world-fixed designs. The screen-fixed HUD was present in all four conditions, and necessary to provide insight into the distance and duration before the next turn (Figure 15). This anticipatory information would otherwise have been absent with the virtual carpet and virtual lead-vehicle, which would be inefficient as drivers would have to drive close to the target before the directional cue is effective. This was also found a downside of both the world-fixed compass concept and the world-fixed pathway concept in the study by Merenda et al. (2018) and by the lead-vehicle studies of Pampel et al. (2019) and Topliss et al. (2019). Providing navigation information using the turn-by-turn concept is indicated to be preferred for HUD design, above showing a map including the route and the current location of the driver (Guo et al., 2014). The HUD information was presented directly in the driver's line of sight, overlaying the road, as this seems to be an ideal display location (Guo et al., 2014; Wittmann et al., 2006). The screen-fixed HUD design includes four elements: element B indicates the current speed limit, element C the current speed of the driver, element D the next navigation maneuver, and element E the distance to this maneuver. The navigation intelligence was created with Unity3D by positioning hidden squares (game objects) at the end of each turn, where the HUD would receive the upcoming navigation information from this square and present it on its display. Because of this, the navigation system did not automatically calculate a new route when the driver would make a navigation error. To solve this, a U-turn was desired which brought the driver back to the route.

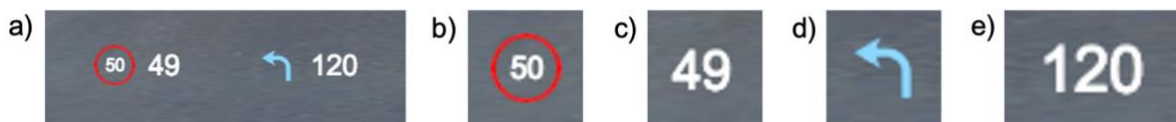


Figure 15. Screen-fixed HUD concept. a) complete concept; b) maximum speed (km/h) indicator; c) current speed (km/h) indicator; d) upcoming turn indicator; e) distance-to-turn (m) indicator

Where the baseline condition only consists of a screen-fixed HUD, the other three conditions additionally include world-fixed navigation information. As explained in Section 2.4.1, world-fixed navigation information is navigation that is attached to specific objects in the driver's field of view, such as the road, or specific points of interest. For concept 1 and 2, the world-fixed navigation information is presented as a virtual carpet. The carpet was created using Sebastian Lague's Bézier Path Creator plug-in (Lague, n.d.), such that first a bézier path was created using anchors and controls after which a mesh was constructed along the path which visualized the carpet. All code used for the path creation can be traced back to Sebastian Lague's Path Creator plug-in. The carpet's material color was set to blue, which is a recommended color for AR automotive HUDs (Merenda et al., 2016). Next to

that, the carpet is slightly transparent such that give-way road markings painted on the road are visible. This was done by setting the rendering mode of the blue carpet material to transparent. The in-game width of the carpet was set to 140 cm which could be done by adjusting the road width provided with the Path Creator plug-in.

Additional to the virtual carpet (basic), the color of the virtual carpet (longitudinal) changed to display speed limit exceedance. The color coding was similar to Jose et al. (2016), however less strict; its color was blue when below the speed limit; yellow when 3% above the speed limit; orange when 6% above the speed limit, and finally it turned red when 9% above the speed limit. The change in color was created to change gradually, meaning that instead of an instant change in color, there is a smooth transition between each color to make the change less sudden and therefore more appealing to use. The C# code used for this color change function can be found in Appendix E (ChangeColor.cs).

Last but not least, for the fourth concept, a virtual lead-vehicle was created. As previous virtual lead-vehicle studies have not clearly mentioned their design decisions for the virtual lead-vehicle, we developed a lead-vehicle based on other literature and our own intuition. First of all, the virtual lead-vehicle is created using a three-dimensional car asset from Green Dino B.V. identical to other cars in the environment. However, only it was changed extensively, such that it would fit our purpose. First of all, all materials of vehicle were created blue, such that it could be distinguished from the other vehicles. Also, as mentioned before, blue is a neutral color and often used for navigation (Merenda et al., 2016). To make the vehicle even more distinguishable from other vehicles, the virtual lead-vehicle was made transparent. This was also important to decrease occlusion of the road information behind the lead-vehicle. Furthermore, the virtual lead-vehicle did not move using the created code from Green Dino B.V., but moved by following the same path as was created for the virtual carpet concepts using Sebastian Lague's Bézier Path Creator plug-in (Lague, n.d.). In addition, the lead-vehicle's behavior consisted of several components. The virtual lead-vehicle was set to drive 1.05 times the speed limit, to make it similarly strict as the longitudinal feedback provided by the virtual carpet (longitudinal). However, when the vehicle would be further than 0.56 times the speed limit away from the driver, the vehicle would brake, and thus wait for the driver to get closer. This is based on the two-second rule, which is a rule-of-thumb normally used to keep a safe distance between a vehicle and its lead-vehicle. This was especially relevant in case of a red traffic light, since the virtual lead-vehicle would drive through the traffic lights, but then waited for the driver as the maximum distance between the lead-vehicle and the driver would be exceeded. The C# code used for the behavior of the virtual lead-vehicle can be found in Appendix E (VCFollowRoute.cs). The virtual lead-vehicle was also designed such that it was not aware of other vehicles on the road, meaning that it drove through them. This was made possible by creating an additional layer and assign it to the virtual lead-vehicle, after which it was set such that the virtual lead-vehicle with the virtual lead-vehicle layer could not collide with game objects with a general vehicle label. Additionally, the virtual lead-vehicle used its indicator lights to indicate whether it was going left or right, such that participants were able to identify the required maneuver at a similar moment as the other conditions (Pampel et al., 2019). Appendix E presents the C# code for the left indicator light, which is highly similar to the C# code for the right indicator light (VCLeftLight.cs). Besides that, the virtual lead-vehicle's brake lights were triggered when the virtual lead-vehicle had to brake (Appendix E, VCbrakelight.cs).

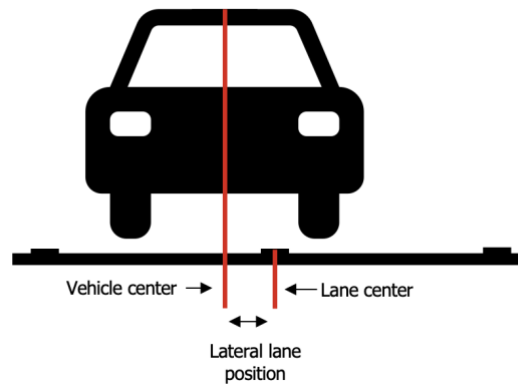
### **3.5 Measurements**

In this study, subjective and objective measures have been performed. The following sections will explain these in more detail.

#### **3.5.1 Objective Measures**

Several measures have been used to assess driving performance, with lateral lane position as the main driving performance measures. The most commonly reported lateral lane position statistic, is lane position variability, indicated with the standard deviation of lateral lane position (Society of

Automotive Engineers, 2012). Especially in driving tasks with a challenging secondary task, the lane position variability is expected to increase compared to a task where the driver is not distracted by a secondary task (Kun, 2018). An increase of lane position variability is a sign of poor driving performance, which can result in an increased probability of collisions (Kun et al., 2009). To obtain the lateral lane position, we measured the distance from the center of the car to the center of the lane the car is driving on (Figure 16).



*Figure 16. Visualization of the lateral lane position*

Additionally, several exploratory objective measures were performed. First of all, the driven speed was measured, which was used to calculate the average speed, and compared with the speed limit to calculate the percentage of speeding during the drive (Tönnis et al., 2007). Both longitudinal measures were interesting to investigate whether the virtual lead-vehicle and the virtual carpet (longitudinal), which both included longitudinal information, affected the speed of the driver. Besides that, the collisions between the participants' vehicle and other vehicles were measured, which is an indicator of driving safety. Although the rate of collisions is low in most driving simulator studies, collisions could not be excluded, which is why it was decided to include the collision rate (Kun, 2018; Medenica et al., 2011). Last but not least, inspired by Pampel et al. (2019) and Topliss et al. (2019), the number of navigation errors were measured, to evaluate the secondary task (i.e., following the navigation instructions) performance.

### **3.5.2 Subjective Measures**

One primary subjective measure and two exploratory subjective measures were performed for each AR HUD navigation concept separately. The driver's mental workload was decided to be the primary objective measure, as it was also measured by the other lead-vehicle studies making it useful for comparison of the results. As mentioned in Section 2.1, secondary tasks like navigation systems demand extra mental resources that are also needed for the primary driving task, which could decrease driving safety (Jensen et al., 2010; Wickens, 2002). In addition, evaluation of drivers' mental workload, in addition to driving errors, is necessary to enhance the effectiveness, usability and acceptability of systems such as navigation systems (Pauzié & Manzano, 2007). To evaluate driver's mental workload, the standardized DALI questionnaire was used (Moray, 1979; Pauzie, 2008). The DALI questionnaire is based on the NASA-TLX workload measurement scale and is designed specifically for the driving context (Pauzie, 2008). The DALI questionnaire was slightly altered for the purpose of this study, by deleting the item concerning Tactile Demand, as no tactile vibrations were present in the driving simulator. This left the DALI questionnaire with the items Global Attention Demand, Visual Demand, Auditory Demand, Stress, Temporal Demand and Interference. Additionally, a few minor adjustments with regard to word usage were performed. The resulting DALI questionnaire used in this study can be found in Appendix A.

As an exploratory subjective measure, perceived safety was measured. Perceived safety is a self-reflective measure that can be defined as the degree to which an individual believes that a system

will affect his or her well-being (Osswald et al., 2012). Perceived safety is critical for the potential use of the system, which is in our case the AR HUD concept (Osswald et al., 2012). To evaluate perceived safety, the perceived safety questionnaire from Cho et al. (2017) was used, which includes three items modified from the Car Technology Acceptance Model, specified for ADAS technologies (Osswald et al., 2012) (Appendix B). This questionnaire was preferred above other perceived safety questionnaires as it only had three items yet shows a proper Cronbach's alpha level ( $\alpha = 0.811$ ).

Thirdly, user experience was evaluated, as this measures both the usability of a product and the emotional connection or feelings of interactive products (Rusu et al., 2015; Schrepp et al., 2014). The shortened user experience (UEQ-S) questionnaire was administered to measure the user experience of the participants when driving with the different AR HUD navigation systems (Schrepp et al., 2017) (Appendix C). The UEQ-S questionnaire was preferred above the original UEQ questionnaire, as the UEQ-short consists of eight instead of 26 items, making the experiment less time consuming. The UEQ-S consists of contrasting attributes, where the first four pairs represent a pragmatic quality scale, and the last four items a hedonic quality scale. This division is based on a theoretical framework of user experience in which the pragmatic quality of a product focusses on the goal oriented or task oriented aspects, and the hedonic quality is based on the non-task oriented quality aspects, such as the originality of the design or the aesthetics of the product (Hassenzahl, 2001; Laugwitz et al., 2008). Participants rated each pair on a 7-point Likert scale, such that it reflected their impression of the AR HUD navigation system.

Finally, participants ranked the four different AR HUD navigation systems based on their perceived safety, perceived driving performance, and perceived mental workload (Appendix D). It might be dubious to include perceived driving performance since it is known that drivers can poorly predict their driving performance (Verster & Roth, 2012). However, these differences in driving performance prediction between participants are nullified by the within-subject design.

### **3.6 Study Procedure**

Throughout the study, several covid-19 measures have been taken. These measures are excluded from the study procedure explanation. In general, they include disinfecting hands before and after entering the experiment room, cleaning all materials, and keeping 1.5-meter distance between the participant and experimenter.

As soon as participants were inside the experiment room, they were asked to sit down behind the desk and read the informed consent, which they signed as they agree upon participation in the study. This was followed by the participant answering several demographic questions on the laptop present on the desk.

Before the experiment started, it was ensured that the participants fully understood its procedure by explaining it multiple times. Firstly, the participant read it in the informed consent form; secondly, the participant read the procedure again on the laptop before answering the demographic questions, and thirdly, a practice trial was run. The practice trial was a 5-minute drive in the environment, without any navigation system present, making sure that errors made in the experiment conditions were not caused by a lack of experience using the driving simulator (Kun, 2018; McGehee et al., 2004). After the participant was invited to take place in the driving simulator, he/she could adjust the seating position such that he/she could comfortably reach the gas and braking pedals. The participants were explained that the practice trial was specifically meant to learn how to use the simulator, and that the simulator could be controlled only by using the gas and breaking pedal, and the steering wheel. Besides that, participants were allowed to use indicator lights if they preferred this, but this was not obligatory. As practice advise (Hock et al., 2018), participants were asked to mainly practice taking a turn, as this was not easy and sometimes participants would get into a "spin". The low chance of simulator sickness was also mentioned such that participants knew that they could stop the experiment at any moment if they experienced any of the symptoms. The advice from Hock et al. (2018) was followed, meaning that not

only in the informed consent form, but also after each trial participants were asked whether they were experiencing any symptoms of motion sickness.

After practicing driving in the simulator, the different AR HUD navigation system designs were presented on paper and explained. This was to avoid that participants had to learn how to use the navigation system concepts during the experiment itself (Faria et al., 2020). When the participant had no more questions, the first trial could be started. For all four trials, participants were presented with a different AR HUD navigation system and had to follow its navigation indications for 7 minutes. After the participant had arrived at the final destination, he/she had to take a seat behind the laptop to evaluate the mental workload, perceived safety, and user experience for that specific navigation system. As soon as all four trials were completed, the ranking questions were presented to the participant. After this, the experiment was over. A short debrief session was provided in which participants were able to ask questions or give other remarks. Additionally, they were given a monetary compensation. The total time per participant was approximately 60 minutes.

### 3.7 Data Processing

#### 3.7.1 Data Gathering

After participants finished the experiment, the driving simulator data was collected and stored into separate files per trial. Each file contained data that was logged per frame. A python script was used to read in the data files and to delete outliers. First, collisions were deleted in case they were not due to bad driving behavior of the participant, but rather due to the traffic's unexpected behavior in the simulator (i.e., a car ignoring a red traffic light). Secondly, data rows were deleted where participants were further away from the center of the lane than 30 meters, excluding all data where participants took a wrong turn. An example of such an occurrence can be found in the location plot of participant 1, trial 1, where the red part of the left plot indicates the wrong route (Figure 17). Additionally, data was deleted manually for cases where participants made more complex route mistakes such as in trial 1 of participant 21 (Figure 17). For such occurrences, the timestamps of these navigation errors were used in the python script to delete the corresponding data rows. Another manual edit was performed for the occurrences where participants had to overtake an unintended traffic congestion. As these encounters resulted in participants driving further away from the center of the lane than otherwise would have been, these values were substituted with their average distance to the center of that lane, which was measured before or after the congestion.

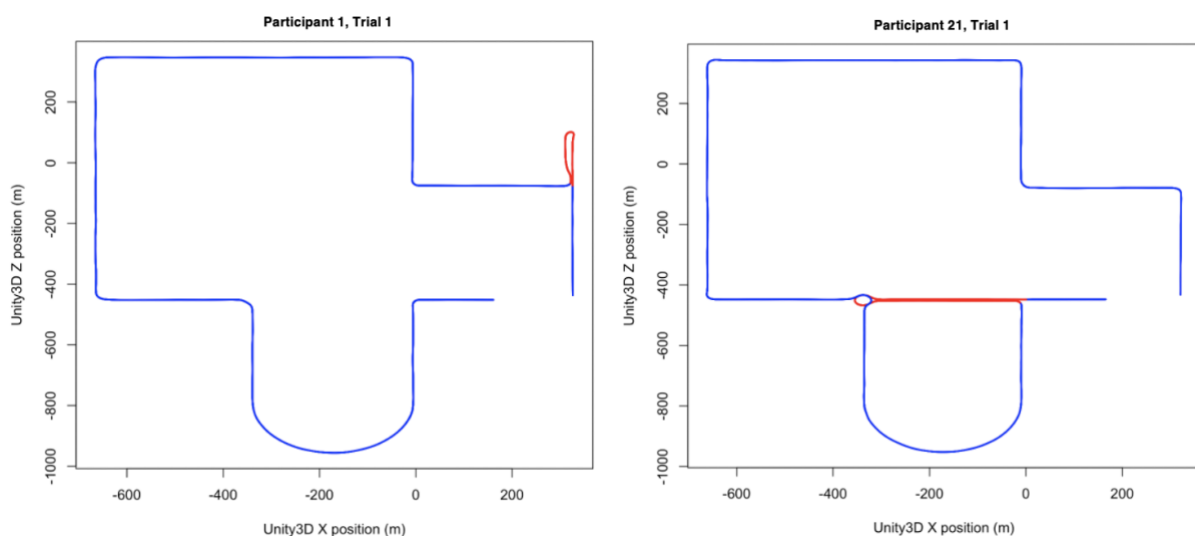


Figure 17. Location plots with navigation errors indicated in red. The left plot shows trial 1 of participant 1. The right plot shows trial 1 of participant 21.

Data modifications were also made for all participants at large. As red traffic light time was not equal in all trials, and speed was close to zero during these instances, it could affect the objective speed analysis. Therefore, all data where the participant's speed was below 0.01 km/h (e.g., they were waiting for a traffic light) was deleted. Additionally, an outlier analysis was performed on all lateral lane position data, to figure out if there were, next to the navigation error occurrences, any additional moments that the lateral lane position value could be considered an outlier. This was expected to be the case, as some participants' vehicles got their car into a spin which caused a deviation of the vehicle speed and an increased lateral lane position. It was found that lateral lane positions higher than 3.024 m could be considered an outlier, as these were larger than the distance between the mean and three times the standard deviation of all lateral lane position data. Therefore, it was decided to delete all data rows where lateral lane position was higher than 3.024 m.

After cleaning the logged data, summarizing variables were created using the Python script, and outputted into a single csv file for data analysis. This file contained four rows of data per participant, with each row representing summarized data of one trial. Besides creating variables over the whole location data per trial, additional variables were created that were either measuring the road parts in the environment, or on the intersection parts. This was done to account for variability of lane position, which can differ depending on the road geometry (Kun, 2018). It can be expected that lane position variability is larger on a curvy road, compared to on a straight road, as more steering is needed.

The responses to all survey questions were automatically saved online. The responses were gathered into one file and exported to a csv file for further analysis. Before performing the data analysis, it was examined whether the questionnaire data was internally consistent by calculating Cronbach's alpha scores. Table 2 shows that each questionnaire includes a good internal consistency, as Cronbach's alpha values are higher than 0.7 (Santos, 1999). The Cronbach's alpha scores are also reported for the individual DALI items, as these were each analyzed separately next to the combined analysis.

Additionally, an outlier analysis was performed per AR HUD concept for each questionnaire's item individually. Values were evaluated being an outlier per variable and AR HUD concept and were removed when they were more than three standard deviations away from the mean. There was no participant that could be considered an outlier for multiple items, therefore only a few outliers for a specific variable have been replaced as a missing value, instead of dropping the data of one participant completely. In total, only 10 values have been replaced as missing. For the demographic variables, participant 30 was considered an outlier for both age (age = 59) and years of driving license (years of license = 40), since both values were more than three times higher than the standard deviation of the participant sample.

Consequently, normality tests were performed for each of the variables, for each AR HUD concept separately. For the driving performance measures all normality was met, except for lateral lane position where it was only met for the virtual carpet (longitudinal) concept. Although not all individual items of the DALI questionnaire were normally distributed, the combined DALI score was normally distributed for each AR HUD concept. In addition, the mean perceived safety was normally distributed as well. For the UEQ-S questionnaire, both the combined UEQ score and the hedonic part of the UEQ questionnaire were normally distributed. However, its pragmatic part was not normally distributed. Similar normality results were found for the ranking variables, where mainly for the virtual lead-vehicle normality was violated.

Table 2

| <i>Internal consistency scores of variables</i> |       |                  |
|-------------------------------------------------|-------|------------------|
| Variable                                        | Items | Cronbach's alpha |
| Mental workload                                 | 6     | 0.77             |
| - Attentional                                   |       | 0.68             |
| - Visual                                        |       | 0.76             |

|                  |   |      |
|------------------|---|------|
| - Auditory       |   | 0.78 |
| - Stress         |   | 0.71 |
| - Temporal       |   | 0.73 |
| - Interference   |   | 0.72 |
| Perceived Safety | 3 | 0.90 |
| User Experience  | 8 | 0.82 |
| - Pragmatic      | 4 | 0.88 |
| - Hedonic        | 4 | 0.91 |

### 3.7.2. Data Analysis

Data was analyzed using STATA 2016 software. Although for some variables normality was not met, it was decided to analyze most of the data using a one-way repeated measures ANOVA, as an ANOVA is robust to departures from normality (Blanca et al., 2017). Data was checked for sphericity to avoid Type 1 errors, but no violations were found (Haverkamp & Beauducel, 2017). In a small proportion of the analyses, when a variable was found to significantly correlate with the dependent variable, it was included in the model as an extra factor (i.e., route, driving frequency and road type) to improve the model (Delaney & Maxwell, 1981). For the covariate age and factors years of license and gender no effects were found at all. Additionally, significant correlations were found between the order of the AR HUD concepts and UEQ-S Pragmatic (Figure 18), visual mental workload, auditory mental workload (Figure 19), and lateral lane position. Three one-way repeated measures ANOVAs were performed, with both the AR HUD concepts and the order of AR HUD concepts as independent variables and either pragmatic user experience, visual mental workload or auditory mental workload as dependent variable. These analyses showed order effects for both UEQ-S Pragmatic,  $F(3, 29) = 3.1$ ,  $p = 0.042$ , and auditory mental workload,  $F(3, 29) = 7.3$ ,  $p < 0.001$ . But, as a latin-square design was used, order effects are averaged out (Voss et al., 2017). Bonferroni corrections were used when performing pairwise comparisons, in order to interpret the significance effects (Armstrong, 2014). In case a variable's normality was rejected Dunn's pairwise comparison test with Bonferroni correction was performed (Dinno, 2015).

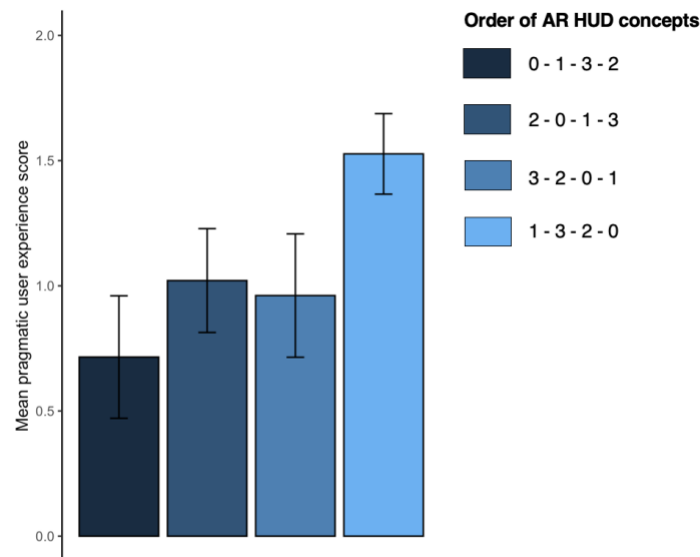
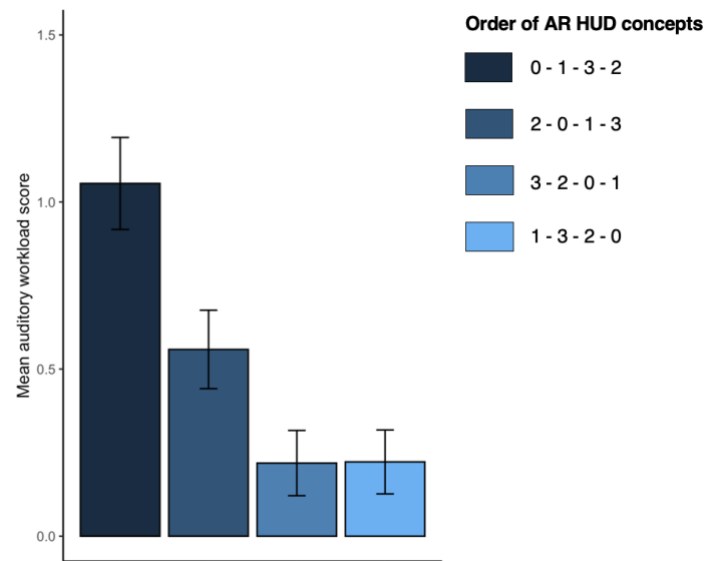


Figure 18. Mean and standard errors (error bars) of the pragmatic user experience score for each order of AR HUD concepts.



*Figure 19.* Mean and standard errors (error bars) of the auditory mental workload score for each order of AR HUD concepts.

## 4. RESULTS

To address the question whether the virtual lead-vehicle AR HUD concept would be better compared to other AR HUD concepts, we will show the effects of the different AR HUD concepts on the differences in means of the dependent variables. Section 4.1 looks into the effect of the AR HUD concepts on lane position variability, after which Section 4.2 looks into the effect of the AR HUD concepts on mental workload. Additionally, the remaining sections cover the exploratory subjective variables (perceived safety, user experience and the ranking data) and exploratory driving performance measures (lateral lane position, average speed and speed limit exceedance). Furthermore, a summary of the mean, standard deviation and number of participants for each variable per AR HUD concept can be found in Appendix F. Throughout this chapter, when mentioning distances, these values are simulated distances which are aimed to be close to physical distances.

### 4.1 Lane Position Variability

Figure 20 shows a higher lane position variability for the virtual carpet (basic) ( $M = 0.23$  m,  $SD = 0.087$  m), compared to the virtual lead-vehicle ( $M = 0.26$  m,  $SD = 0.062$  m). It was, however, expected that the virtual lead-vehicle would lead to a lower lane position variability compared to the virtual carpet (H1). However, although a one-way repeated measures ANOVA revealed significant differences for the mean lane position variability between AR HUD concepts,  $F(3,94) = 8.3$ ,  $p < 0.01$ , pairwise comparisons with Bonferroni correction indicate that if there would be an opposite effect between the virtual carpet (basic) and the virtual lead-vehicle on lane position variability, this effect is not significant ( $\Delta M = 0.03$  m,  $p = 0.88$ ).

In addition, it was expected that the addition of longitudinal information to the navigation information leads to a lower lane position variability (H3). However, looking at Figure 20, no difference in lane position variability can be observed between the virtual carpet (basic) and virtual carpet (longitudinal) concepts. A Bonferroni corrected pairwise comparison of the virtual carpet (basic) ( $M = 0.23$  m,  $SD = 0.087$  m) with the virtual carpet (longitudinal) ( $M = 0.23$  m,  $SD = 0.070$  m) also does not show a significant difference between these concepts ( $\Delta M = 0.003$  m,  $p = 1.00$ ).

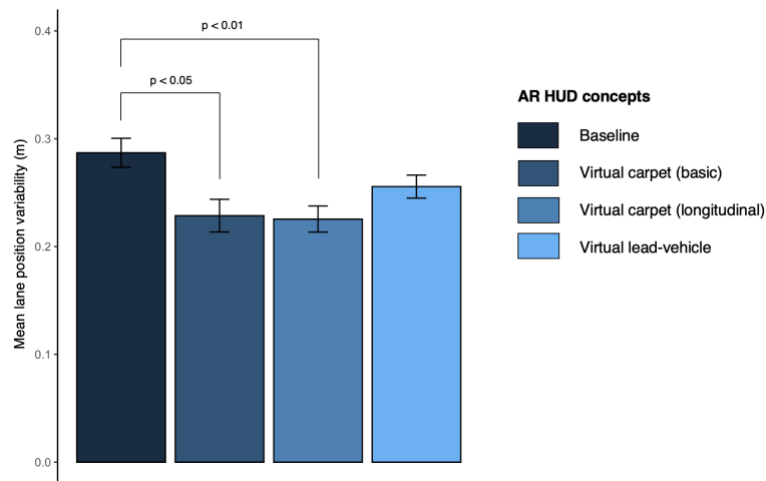


Figure 20. Means and standard errors (error bars) of lane position variability for all AR HUD concepts. Significant interactions are shown when  $p < 0.05$ , when  $p < 0.01$ , and when  $p < 0.001$ .

#### 4.1.1 Effect of Road

Figure 21 shows the difference in mean lane position variability for participants driving on an intersection, compared to participants driving on the road. From Figure 21 it can be observed that lane position variability is much higher for the intersection fragments compared to the road fragments. A

one-way repeated measures ANOVA including both road type and AR HUD concepts as independent variables and lane position variability as the dependent variable, shows that indeed the effect of road fragments on lane position variability is significant,  $F(1, 32) = 113, p < 0.001$ . In addition, the effect of the AR HUD concepts on lane position variability score remains of similar magnitude,  $F(3,192) = 8.1, p < 0.001$ . However, for both the road fragments ( $\Delta M = 0.03$  m,  $p = 0.50$ ) and intersection fragments ( $\Delta M = 0.06$  m,  $p = 1.00$ ) the difference between the virtual lead-vehicle and the virtual carpet (basic) is still not significant. This is similar for the comparisons of the virtual carpet (basic) with the virtual carpet (longitudinal) for both the road ( $\Delta M = 0.007$  m,  $p = 1.00$ ) and intersection ( $\Delta M = 0.007$  m,  $p = 1.00$ ) fragments.

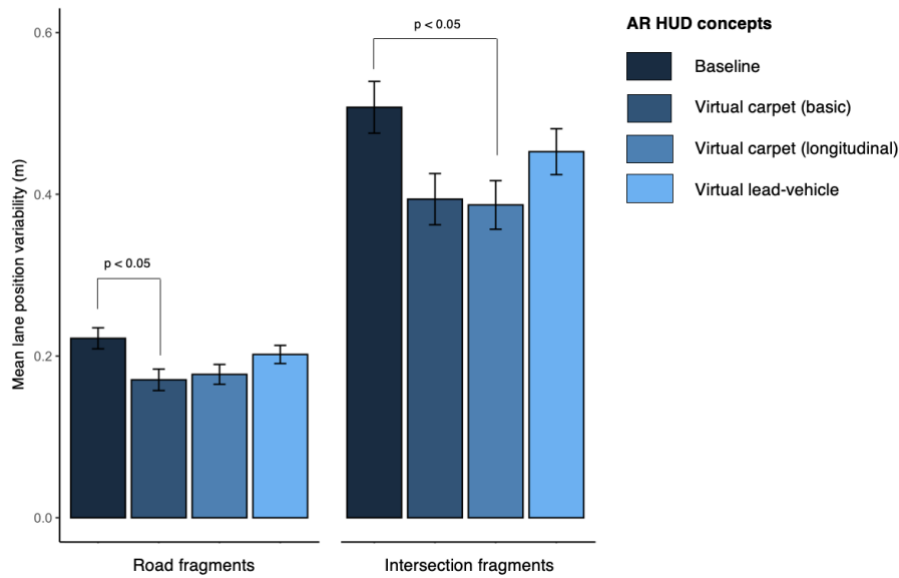


Figure 21. Means and standard errors (error bars) of lane position variability for all AR HUD concepts for road fragments and intersection fragments. Significant interactions are shown when  $p < 0.05$ , when  $p < 0.01$ , and when  $p < 0.001$ .

## 4.2 Mental Workload

To evaluate the effect of the AR HUD concepts on mental workload, not only the combined DALI score, but also the individual DALI items are evaluated (Figure 22). This is the recommended approach when individual weights are not calculated for each DALI item (Hart, 2006).

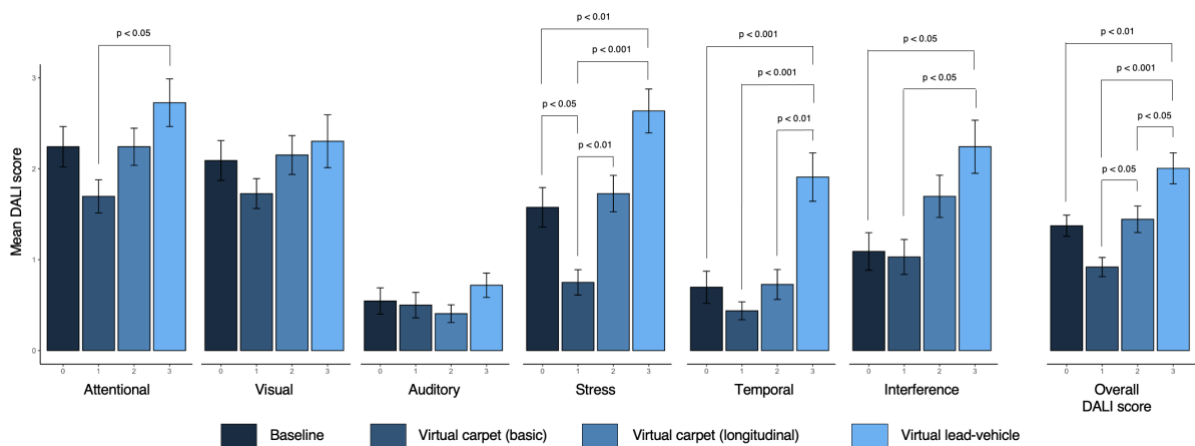


Figure 22. Means and standard errors (error bars) of mental workload per AR HUD condition for each DALI item individually and a combined DALI score. Significant interactions are shown when  $p < 0.05$ , when  $p < 0.01$ , and when  $p < 0.001$ .

Mental workload was presented as a dependent variable in both hypothesis 2 and 4. First of all, it was hypothesized that, compared to the virtual carpet (basic), the virtual lead-vehicle leads to a lower mental workload (H2). Secondly, it was expected that adding longitudinal information to the AR HUD navigation concept leads to a decreased mental workload (H4). However, looking at the overall mental workload and its separate items, the virtual lead-vehicle has a higher mental workload compared to the virtual carpet (basic). In addition, longitudinal information seems to increase mental workload instead of decrease.

A one-way repeated measures ANOVA with AR HUD concepts as independent variable and overall mental workload as the dependent variable revealed that there is a significant effect of the AR HUD concepts on the overall mental workload,  $F(3, 96) = 14$ ,  $p < 0.001$ . In addition, six separate one-way repeated measures ANOVAs were run, each with one of the mental workload items as the dependent variable, and AR HUD concepts as the independent variable. These analyses showed significant effects of the AR HUD concepts on attentional mental workload,  $F(3, 96) = 4.5$ ,  $p = 0.006$ , stress,  $F(3, 96) = 16$ ,  $p < 0.001$ , temporal mental workload,  $F(3, 95) = 18$ ,  $p < 0.001$ , and interference,  $F(3, 96) = 6.5$ ,  $p < 0.001$ , but not on visual,  $F(3, 96) = 1.3$ ,  $p = 0.29$ , and auditory, ( $F(3, 96) = 2.2$ ,  $p = 0.09$ , mental workload.

Pairwise comparisons with Bonferroni correction shows that when comparing the overall mental workload scores of the virtual carpet (basic) ( $M = 0.92$ ,  $SD = 0.60$ ) with the virtual lead-vehicle ( $M = 2.0$ ,  $SD = 0.98$ ) that mental workload caused by the virtual lead-vehicle is significantly higher than for the virtual carpet (basic) ( $\Delta M = 1.1$ ,  $p < 0.001$ ). In fact, Figure 22 shows that when looking at the other two AR HUD concepts, that mental workload is highest for the virtual lead-vehicle and lowest for the virtual carpet (basic). Additionally, Dunn's pairwise comparisons with Bonferroni correction show that the higher mental workload for the virtual lead-vehicle compared to the virtual carpet (basic) is also found to be significant for the attentional mental workload ( $z = 3.3$ ,  $p = 0.0028$ ), stress ( $z = 5.5$ ,  $p < 0.001$ ), temporal mental workload ( $z = 4.2$ ,  $p < 0.001$ ) and interference ( $z = 2.8$ ,  $p = 0.014$ ), but not for the visual ( $z = 2.0$ ,  $p = 0.125$ ) and auditory mental workload ( $z = 1.5$ ,  $p = 0.340$ ). Altogether, it is found that hypothesis 2 can be rejected, and that an opposite effect might be true.

For evaluating the addition of longitudinal information to the virtual carpet (basic), the means of the virtual carpet (basic) and the virtual carpet (longitudinal) are compared. Figure 22 shows that for all separate mental workload items mental workload is higher when longitudinal information was added to the virtual carpet (basic), except for auditory mental workload. Therefore, the hypothesis that mental workload would be lower when longitudinal information would be added to the virtual carpet (basic) can be rejected. An opposite effect might be true, which is significant for the overall mental workload (Bonferroni corrected pairwise comparison,  $\Delta M = 0.53$ ,  $p = 0.046$ ) and for stress (Dunn's pairwise comparison,  $z = 3.1$ ,  $p = 0.005$ ), but we have to be careful concluding this as not all mental workload items indicate this.

Looking at Figure 22, further analysis on the individual items of mental workload shows that the virtual lead-vehicle not only has a higher mental workload compared to the virtual carpet (basic), but it even has a higher mental workload compared to all three AR HUD concepts. Dunn's pairwise comparisons with Bonferroni correction show that only the temporal mental workload score is significantly highest for the virtual lead-vehicle compared to the other three concepts (baseline:  $z = 3.6$ ,  $p < 0.001$ ; virtual carpet (basic):  $z = 4.2$ ,  $p < 0.001$ ; virtual carpet (longitudinal):  $z = 3.3$ ,  $p = 0.003$ ).

Furthermore, when comparing the virtual lead-vehicle's mental workload with the baseline AR HUD concept's mental workload, Dunn's pairwise comparison tests with Bonferroni adjustments show that the virtual lead-vehicle's mental workload is significantly higher for stress ( $z = 3.0$ ,  $p = 0.008$ ), temporal mental workload ( $z = 3.6$ ,  $p < 0.001$ ) and interference ( $z = 2.8$ ,  $p = 0.016$ ).

Additionally, the virtual carpet (basic) shows the lowest mental workload scores for all mental workload items and the overall mental workload item, except the auditory mental workload. Dunn's pairwise comparisons with Bonferroni correction show that only the stress score is significantly lowest

for the virtual carpet (basic) compared to the other three concepts (baseline:  $z = 2.6$ ,  $p = 0.031$ ; virtual carpet (longitudinal):  $z = 3.1$ ,  $p = 0.005$ ; virtual lead-vehicle:  $2.4$ ,  $p = 0.046$ ).

These additional analyses on mental workload can conclude that overall, the virtual carpet (basic) performs best with regard to mental workload, i.e., it has the lowest mental workload score, and the virtual lead-vehicle scores worst with regard to mental workload, i.e., it has the highest mental workload.

#### 4.2.1 Effects of Driving Frequency and Route

As explained in Section 3.7.2, for some one-way repeated ANOVA analyses the variables route and driving frequency were included as an additional factor since they improved the model.

First of all, looking at Figure 23, it can be seen that the overall mental workload is higher for route A compared to route B. A one-way repeated measures ANOVA with independent variables AR HUD concepts and route, and dependent variable overall mental workload, showed that indeed a significant effect exists between route and overall mental workload,  $F(1, 32) = 9.8$ ,  $p = 0.004$ . Not surprisingly, the effect of AR HUD concepts on the combined DALI score decreases when controlling for route ( $F(3, 63) = 7.3$ ,  $p < 0.001$ ). However, pairwise comparisons with Bonferroni adjustments still found significant differences between the virtual carpet (basic) and the virtual lead-vehicle for both route A ( $\Delta M = 1.2$ ,  $p < 0.001$ ) and route B ( $\Delta M = 0.91$ ,  $p = 0.013$ ). However, the same pairwise comparisons show that both for route A ( $\Delta M = 0.50$ ,  $p = 0.371$ ) and route B ( $\Delta M = 0.55$ ,  $p = 0.322$ ) there is no difference in lane position variability between the virtual carpet (basic) and virtual carpet (longitudinal), suggesting that making a conclusion about the addition of longitudinal information should be done with even more care.

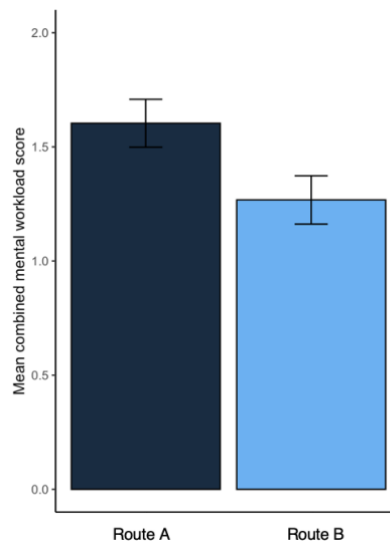


Figure 23. Means and standard errors (error bars) of the combined mental workload score for both route A and route B.

Additionally, both route (Figure 24) and driving frequency (Figure 25) significantly affect the interference score next to AR HUD concepts. Figure 24 shows that route A leads to a higher interference score compared to route B. Additionally, participants who drove at least once a week in the last three months gave a higher interference score compared to participants who drove once a month or less. One-way repeated measures ANOVAs have been performed with AR HUD concept as independent variable and either route or driving frequency as a second independent variable and interference as the dependent variable. These analyses found that indeed route A leads to a significantly higher interference compared to route B,  $F(1,32) = 14$ ,  $p < 0.001$ . Similarly, Figure 25 shows that participants who drive more often gave significantly higher interference scores,  $F(6, 26) = 3$ ,  $p = 0.022$ . Where the effect of AR HUD

concepts on the DALI Interference score decreases when controlling for route ( $F(3, 63) = 4.1, p = 0.011$ ), the effect stays similar when controlling for driving frequency ( $F(3, 96) = 6.5, p < 0.001$ ). Interestingly, Dunn's pairwise comparison test with Bonferroni adjustments indicates that the difference between the virtual carpet (basic) and the virtual lead-vehicle, explaining hypothesis 2, only remains significant for route A ( $z = 3.4, p = 0.002$ ), but not for route B ( $z = 0.58, p = 1.00$ ). Furthermore, the same pairwise comparisons show that the difference between the baseline concept and the virtual lead-vehicle is no longer significant on either route A ( $z = 2.3, p = 0.063$ ) or route B ( $z = 1.8, p = 0.198$ ). Besides that, the pairwise comparisons also show that the difference between the baseline concept and the virtual carpet (longitudinal) does become slightly significant for route B ( $z = 2.5, p = 0.043$ ). For participants that indicated that they drove 2-3 times per month (i.e., driving frequency = 3), Dunn's pairwise comparison test with Bonferroni adjustments shows a significant difference between the baseline concept and the virtual lead-vehicle ( $z = 2.9, p = 0.012$ ). No other significant effects were found for the contrast variable driving frequency.

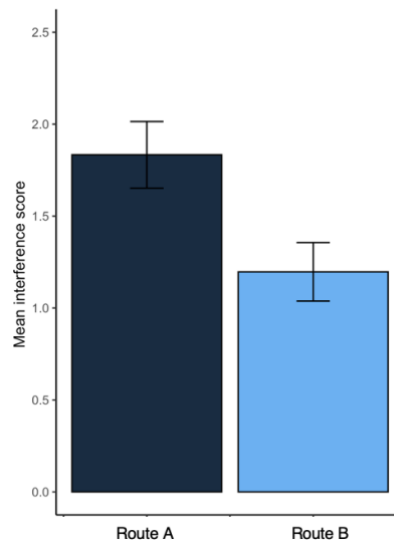


Figure 24. Means and standard errors (error bars) of the interference score for both route A and route B.

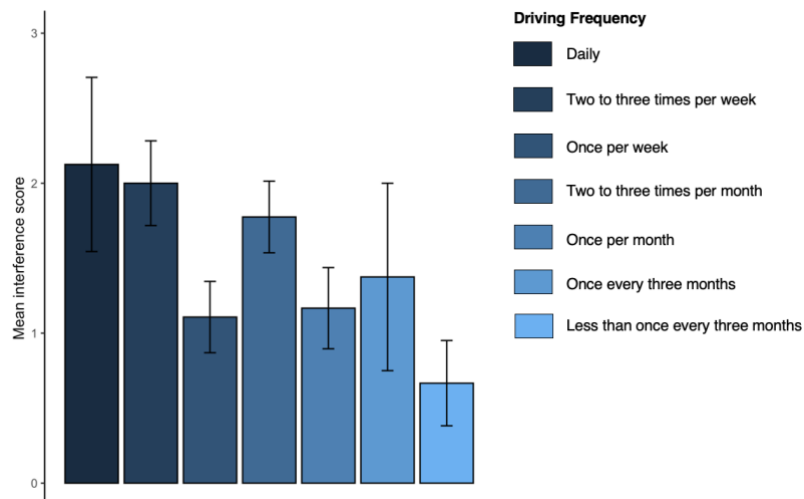


Figure 25. Means and standard errors (error bars) of interference for different levels of driving frequency.

### 4.3 Further Data Analysis

Next to the analysis on mental workload and lane position variability, we will now explore the results with respect to the exploratory variables. First, findings with respect to the exploratory driving performance measures will be given, including findings of lane position variability other than our primary aim. Second, the subjective exploratory findings are presented. As these measures are exploratory, findings will not be compared with hypotheses when looking at these results.

#### 4.3.1 Exploratory Driving Performance Measures

As exploratory driving performance measures, lane position variability will be first analyzed for further findings. Next, lateral lane position, average speed, speed limit exceedance, navigation errors and the number of collisions will be discussed.

Although no significant differences in lane position variability were found with respect to our primary objective, Figure 26 shows that the baseline concept has higher lane position variability compared to all three other AR HUD concepts. When analyzing the lane position variability data for other effects using Bonferroni corrected pairwise comparisons, it is found that the baseline concept leads to a significantly higher lane position variability compared to the virtual carpet (basic) ( $\Delta M = 0.06$  m,  $p = 0.011$ ) and the virtual carpet (longitudinal) ( $\Delta M = 0.06$  m,  $p = 0.007$ ), but not compared to the virtual lead-vehicle ( $\Delta M = 0.03$  m,  $p = 0.555$ ). However, when performing Bonferroni corrected pairwise comparisons on the road data shows that, only the virtual carpet (basic) has significantly lower lane position variability compared to the baseline concept ( $\Delta M = 0.05$  m,  $p = 0.028$ ). Conversely, although the differences seem larger for the intersection data compared to the overall lane position variability data, pairwise comparisons with Bonferroni correction show that only the difference between the baseline concept and the virtual carpet (longitudinal) is significant ( $\Delta M = 0.12$  m,  $p = 0.036$ ).

Furthermore, Figure 26 shows that, similar to lane position variability, the baseline concept caused participants to drive furthest away from the center of the lane, followed by the virtual lead-vehicle, the virtual carpet (longitudinal) and the virtual carpet (basic). A one-way repeated measures ANOVA with AR HUD concepts as the independent variable and lateral lane position as the dependent variable found that the lateral lane position is significantly different in relation to the AR HUD concepts,  $F(3, 94) = 15$ ,  $p < 0.001$ . A pairwise post-hoc Dunn test with Bonferroni adjustments indicates that the baseline concept has a significantly larger lateral lane position compared to the virtual carpet (basic) ( $z = 2.6$ ,  $p = 0.035$ ) and virtual carpet (longitudinal) ( $z = 2.6$ ,  $p = 0.025$ ) concepts, but not compared to the virtual lead-vehicle ( $z = 1.2$ ,  $p = 0.685$ ). Adding road type as an additional factor next to AR HUD concepts to the one-way repeated measures ANOVA showed an effect of road type on lateral lane position,  $F(1, 32) = 113$ ,  $p < 0.001$ . The effect of AR HUD concepts on lateral lane position remained present, but decreased,  $F(3, 96) = 10$ ,  $p < 0.001$ . Dunn's pairwise comparison test with Bonferroni adjustments only revealed a significant difference on the road data between the baseline concept and the virtual carpet (longitudinal) ( $z = 2.6$ ,  $p = 0.028$ ).

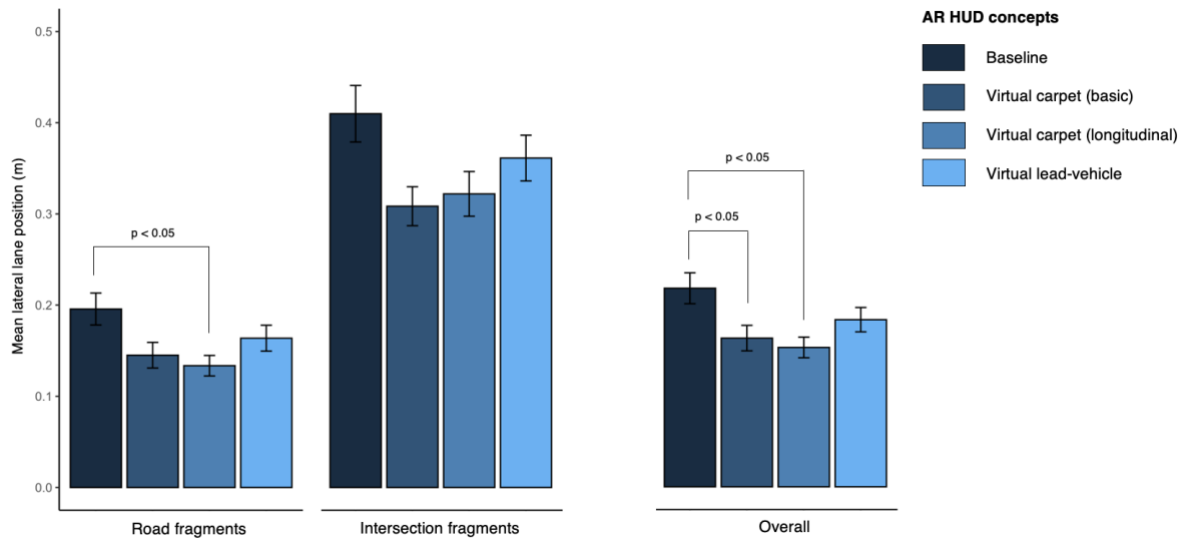


Figure 26. Means and standard errors (error bars) of the lateral lane position (m) for all AR HUD concepts. Results are presented on the road data, intersection data, and overall data. Significant interactions are shown when  $p < 0.05$ , when  $p < 0.01$ , and when  $p < 0.001$ .

As a longitudinal measure, the driver's speed was measured during each trial. Figure 27 shows that the average speed is found to be very similar for all AR HUD concepts, ranging between 43.3 km/h for the baseline concept to 43.9 km/h for the virtual carpet (basic) concept. A one-way repeated measures ANOVA with AR HUD concepts as the independent variable and average speed as the independent variable shows that no significant difference is found between the different AR HUD concepts for the average speed,  $F(3, 96) = 0.57$ ,  $p = 0.63$ . Additionally, Figure 27 shows that the average speed is lower for the intersection fragments compared to the road fragments. Adding road type to the one-way repeated measures ANOVA as an extra independent variable shows this effect of road type on average speed is statistically significant,  $F(1, 32) = 1493$ ,  $p < 0.001$ . It also slightly improves the effect of AR HUD concepts on average speed, but it is even less significant,  $F(3, 96) = 1.4$ ,  $p = 0.244$ .

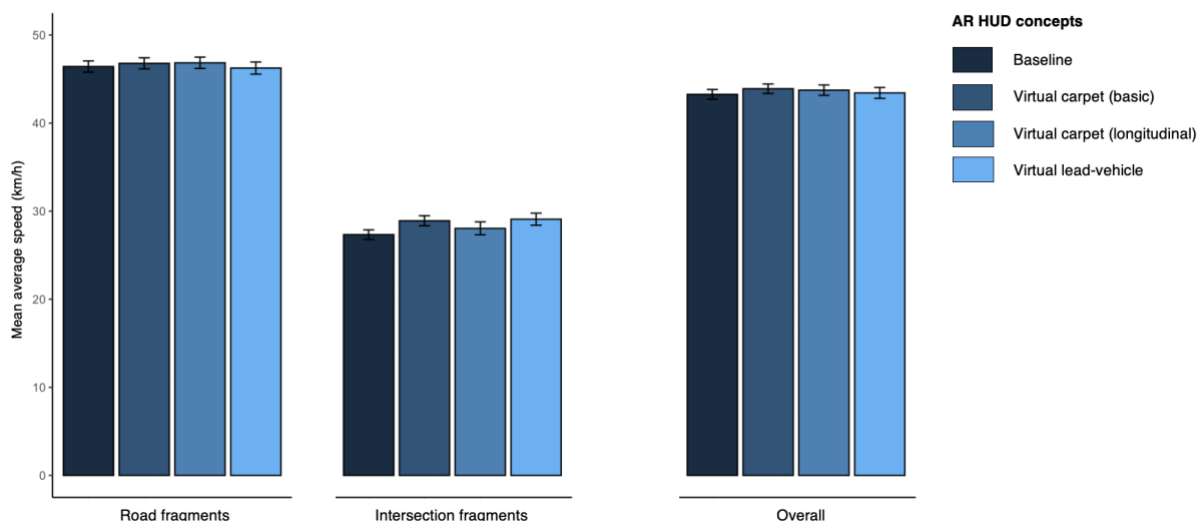


Figure 27. Means and standard errors (error bars) of average speed (km/h) for all AR HUD concepts. Results are presented on the road data, intersection data, and overall data. Significant interactions are shown when  $p < 0.05$ , when  $p < 0.01$ , and when  $p < 0.001$ .

Another variant of the speed data was created by calculating the percentage of the timeframes where the driver was exceeding the speed limit, named speed limit exceedance. Figure 28 shows that with the virtual car (basic) concept, participants exceeded the speed limit most often ( $M = 28\%$ ,  $SD = 9.7\%$ ), followed by the baseline ( $M = 27\%$ ,  $SD = 11\%$ ), the virtual carpet (longitudinal) ( $M = 23\%$ ,  $SD = 11\%$ ), and the least speed limit exceedance was found with the virtual lead-vehicle ( $M = 21\%$ ,  $SD = 9.5\%$ ). The overall effect of speed limit exceedance was found to be statistically significant,  $F(3,96) = 12.6$ ,  $p < 0.001$ . Pairwise comparisons with Bonferroni adjustments show that the virtual lead-vehicle leads to significantly lower speed limit exceedance compared to the virtual carpet (basic) ( $\Delta M = 7.1\%$ ,  $p = 0.031$ ). Figure 28 also shows that the speed limit is exceeded more often on the road fragments, compared to on the intersection fragments. When adding road type as an additional independent variable to the one-way repeated measures ANOVA with AR HUD concepts as the other independent variable and speed limit exceedance as the dependent variable, road type significantly influences speed limit exceedance,  $F(1, 32) = 181$ ,  $p < 0.001$ . Additionally, the effect of AR HUD concepts remains, but is a bit lower,  $F(3, 96) = 8.3$ ,  $p < 0.001$ . Bonferroni corrected pairwise comparisons show that the significant difference in speed exceedance between the virtual carpet (basic) and the virtual lead-vehicle is only significant on the road fragments ( $\Delta M = 8.4\%$ ,  $p = 0.021$ ).

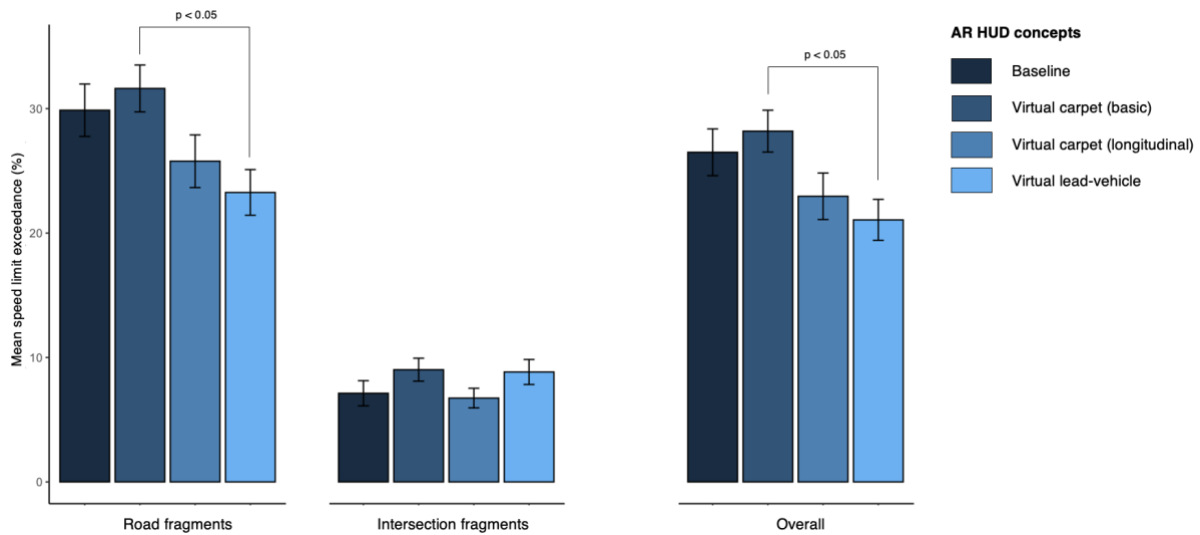


Figure 28. Means and standard errors (error bars) of speed limit exceedance (%) for all AR HUD concepts. Results are presented on the road data, intersection data, and overall data. Significant interactions are shown when  $p < 0.05$ , when  $p < 0.01$ , and when  $p < 0.001$ .

Additionally, navigation errors were measured, indicating when the driver made a mistake when following the navigation instructions. Table 3 shows that for the baseline AR HUD concept more navigation errors were made compared to the other concepts. A one-way repeated measures ANOVA with AR HUD concepts as the independent variable and navigation errors as the dependent variable indicated that the effect of navigation errors on AR HUD concepts is statistically,  $F(3, 96) = 7.4$ ,  $p < 0.001$ . Additionally, Dunn's pairwise comparison test with Bonferroni adjustments shows that indeed the baseline concept resulted in a higher number of navigation errors compared to all three other AR HUD concepts (virtual carpet (basic):  $z = 4.0$ ,  $p < 0.001$ ; virtual carpet (longitudinal):  $z = 0.4$ ,  $p < 0.001$ ; virtual lead-vehicle:  $z = 3.1$ ,  $p = 0.005$ ). Interestingly, six of the fifteen navigation errors occurred at the first intersection of route A, followed, by three errors at intersection 7, two errors at the start of the drive, and at intersection 3, 4, 6 and 11 all occurred one navigation error (see Figure 14b to localize the intersections).

Furthermore, only four collisions were observed in all 132 trials, namely three for the virtual carpet (basic) and one for the virtual lead-vehicle. This low number of collisions is not surprising, as in similar driving simulator studies collisions did not occur a lot either (Kun, 2018; Medenica et al., 2011). Additionally, a one-way repeated measures ANOVA with AR HUD concepts as the independent

variable and collisions as the dependent variable showed that this difference is not statistically significant,  $F(3, 96) = 2.1$ ,  $p = 0.110$ .

Table 3

*Number of collisions and navigation errors per AR HUD concept.*

| Variable          | Baseline | Virtual carpet (basic) | Virtual carpet (longitudinal) | Virtual lead-vehicle |
|-------------------|----------|------------------------|-------------------------------|----------------------|
| Navigation errors | 13       | 0                      | 0                             | 2                    |
| Collisions        | 0        | 3                      | 0                             | 1                    |

#### 4.3.2 Exploratory Subjective Measures

As subjective measures, perceived safety, user experience and ranking data on perceived driving performance, perceived safety and perceived mental workload were collected.

Looking at Figure 29, it can be clearly seen that the virtual lead-vehicle was perceived as less safe compared to the other three AR HUD concepts. Additionally, no strong difference in perceived safety seems to be present between the baseline, virtual carpet (basic) and virtual carpet (longitudinal) concepts. A repeated measures ANOVA with AR HUD concepts as the independent variable and perceived safety as the dependent variable showed that, indeed, there is a significant effect of the AR HUD concepts on perceived safety,  $F(3, 96) = 79$ ,  $p < 0.001$ . Our observations are also supported by pairwise comparisons, as pairwise comparisons with Bonferroni correction indicate that the virtual lead-vehicle scores significantly worse on perceived safety compared to the baseline concept, the virtual carpet (basic) and the virtual carpet (longitudinal) ( $\Delta M = 3.5$ ,  $p < 0.001$ ;  $\Delta M = 3.1$ ,  $p < 0.001$ ;  $\Delta M = 2.9$ ,  $p < 0.001$ ). Additionally, the pairwise comparisons indeed do not show significant differences in perceived safety between the baseline concept, the virtual carpet (basic) and the virtual carpet (longitudinal).

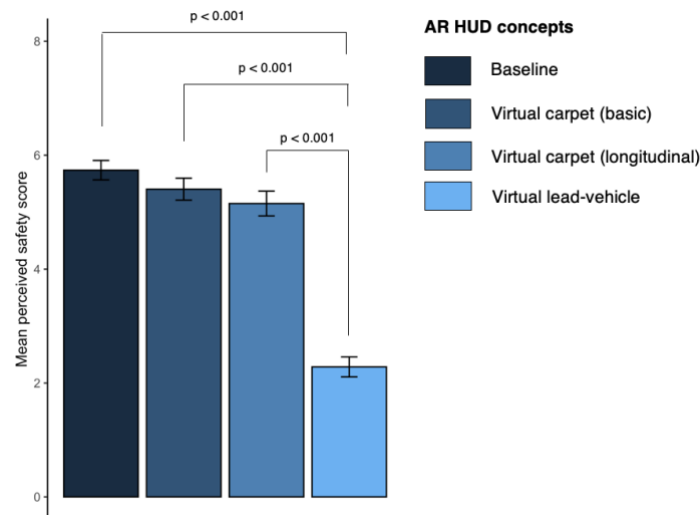


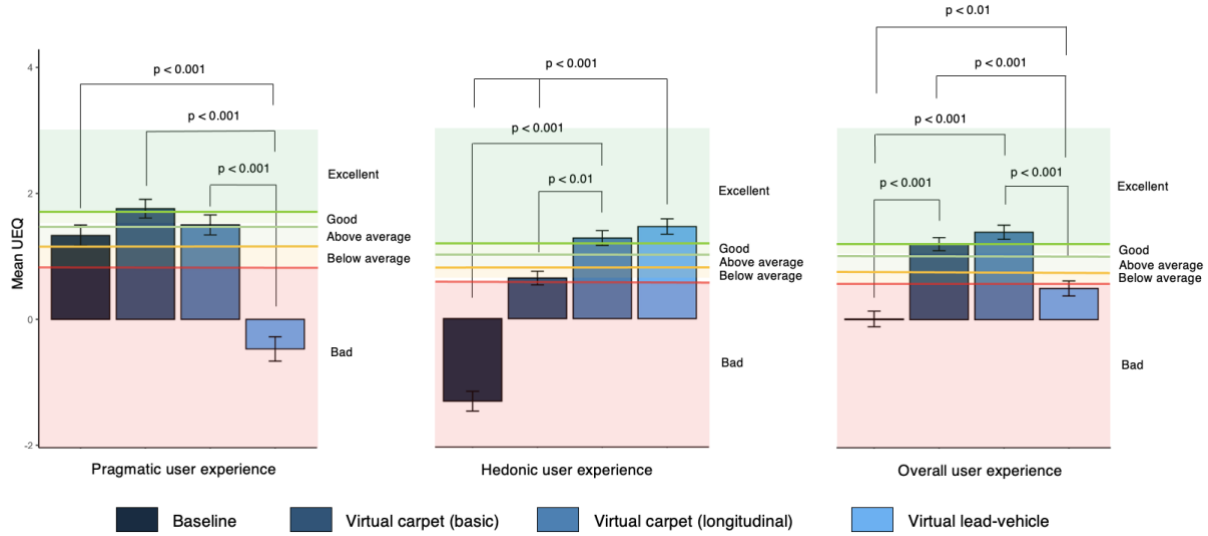
Figure 29. Means and standard errors (error bars) of perceived safety for all AR HUD concepts. Significant interactions are shown when  $p < 0.05$ , when  $p < 0.01$ , and when  $p < 0.001$ .

Figure 30 shows the means and standard errors on user experience per AR HUD concept, separated into both pragmatic user experience and hedonic user experience and provided as overall user experience. The benchmarks retrieved from Schrepp et al. (2017) are indicated using colors, such that the red area indicated a bad user experience, the yellow area a user experience that is above average, and different shades of green indicate average, good and excellent user experience.

Looking at the overall user experience, it can be seen that both the baseline AR HUD concept and the virtual lead-vehicle have lower user experience compared to the two virtual carpet concepts. The overall user experience for the baseline concept and virtual lead-vehicle are even benchmarked as bad, where the virtual carpet concepts show good and excellent overall user experience. When performing a one-way repeated measures ANOVA, a significant effect of the AR HUD concepts on user experience is found,  $F(3, 96) = 35, p < 0.001$ . Pairwise comparisons using a Bonferroni correction match our observations as they indicate a significant lower user experience for the virtual lead-vehicle compared to the virtual carpet (basic) ( $\Delta M = 0.70, p < 0.001$ ) and virtual carpet (longitudinal) ( $\Delta M = 0.89, p < 0.001$ ). Similarly, pairwise comparisons using a Bonferroni correction also show significant lower user experience for the baseline concept compared to the virtual carpet (basic) ( $\Delta M = 1.2, p < 0.001$ ) and the virtual carpet (longitudinal) ( $\Delta M = 1.4, p < 0.001$ ). In addition, the baseline concept results in an even lower overall user experience compared to the virtual lead-vehicle ( $\Delta M = 0.48, p = 0.021$ ). Finally, when comparing the two virtual carpet concepts, no statistical difference on overall user experience can be found ( $\Delta M = 0.19, p = 1.00$ ).

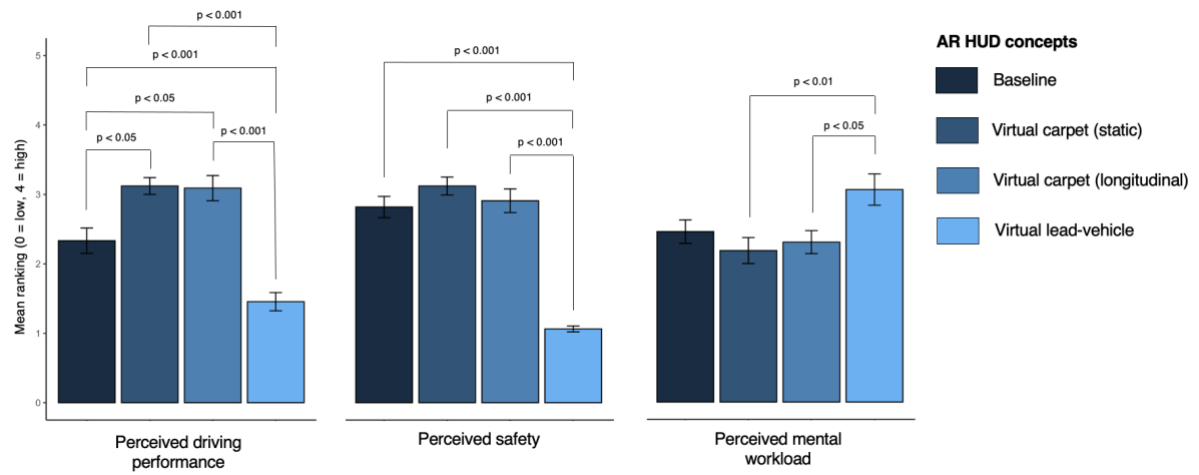
Furthermore, when looking at Figure 30 it can be seen that the results of the hedonic items and pragmatic items are very opposite to each other, which is why we analyze them separately as well. For pragmatic user experience, it can be found that the virtual lead-vehicle has a much lower pragmatic user experience compared to the other three AR HUD concepts. Considering the benchmarks, the virtual lead-vehicle clearly has a bad user experience, where all other three AR HUD concepts score at least above average for pragmatic user experience, with the virtual carpet (basic) having an excellent pragmatic user experience. A one-way repeated measures ANOVA with AR HUD concepts as the independent variable and pragmatic user experience as the dependent variable shows a significant effect of the AR HUD concepts on pragmatic user experience,  $F(3, 96) = 41, p < 0.001$ . Dunn's pairwise comparison test with Bonferroni adjustments shows that the virtual lead-vehicle has a lower pragmatic user experience compared to the baseline concept, the virtual carpet (basic) and the virtual carpet (longitudinal) ( $z = 5.0, p < 0.001$ ;  $z = 6.6, p < 0.001$ ;  $z = 5.4, p < 0.001$ ).

Additionally, considering the hedonic user experience, opposite results are found. For the hedonic user experience, it can be observed that the baseline concept scores the lowest on hedonic user experience, which is benchmarked as bad hedonic user experience, followed by the virtual carpet (basic) with below average hedonic user experience, and the virtual carpet (longitudinal) and virtual lead-vehicle with both excellent hedonic user experience. A one-way repeated measures ANOVA shows that there is a significant effect of AR HUD concepts on the hedonic user experience,  $F(3, 96) = 109, p < 0.001$ . All pairwise comparisons with Bonferroni correction are significant except the comparison between the virtual lead-vehicle and the virtual carpet (longitudinal) ( $\Delta M = 0.18, p = 1.00$ ).



**Figure 30.** Means and standard errors (error bars) of the user experience scores for all AR HUD concepts. Significant interactions are shown when  $p < 0.05$ , when  $p < 0.01$ , and when  $p < 0.001$ . Benchmarks are shown with red (bad), yellow (below average) and green (above average, good, and excellent) colors.

Finally, after the participants performed all four trials, they were asked to rank the AR HUD concepts based on their perceived driving performance, perceived safety and perceived mental workload. The mean ranking per concept is visualized in Figure 31. From Figure 31 it can be seen that perceived driving performance is ranked best for both the virtual carpet concepts, and worst for the virtual lead-vehicle. In addition, similar to the perceived safety measure, the ranking data shows that the virtual lead-vehicle is ranking much lower with respect to perceived safety compared to the other three AR HUD concepts. With respect to the ranking of perceived mental workload, the virtual lead-vehicle is ranked as needing the most perceived mental workload compared to the other three AR HUD concepts. For all three ranking variables a significant effect is found of the AR HUD concepts on their ranking when using a one-way repeated measures ANOVA (perceived driving performance:  $F(3, 96) = 19$ ,  $p < 0.001$ ; perceived safety:  $F(3, 96) = 39$ ,  $p < 0.001$ ; perceived mental workload:  $F(3, 96) = 3.2$ ,  $p = 0.026$ ). However, the effect of mental workload is rather small. Dunn's pairwise comparison test with Bonferroni adjustments shows that perceived driving performance is significantly lower for the virtual lead-vehicle compared to the other AR HUD concepts (baseline:  $z = 3.2$ ,  $p = 0.004$ ; virtual carpet (basic):  $z = 6.0$ ,  $p < 0.001$ ; virtual carpet (longitudinal):  $z = 5.9$ ,  $p < 0.001$ ). Similarly, Dunn's pairwise comparisons with Bonferroni adjustments show that the virtual lead-vehicle is ranked with a significantly lower perceived safety score compared to the other three AR HUD concepts (baseline:  $z = 6.3$ ,  $p < 0.001$ ; virtual carpet (basic):  $z = 7.4$ ,  $p < 0.001$ ; virtual carpet (longitudinal):  $z = 6.7$ ,  $p < 0.001$ ). Additionally, Dunn's pairwise comparison test with Bonferroni adjustments shows that perceived mental workload is indeed ranked significantly higher for the virtual lead-vehicle compared to the virtual carpet (basic) ( $z = 3.2$ ,  $p = 0.004$ ) and virtual carpet (longitudinal) ( $z = 2.7$ ,  $p = 0.018$ ), but not compared to the baseline concept ( $z = 2.2$ ,  $p = 0.085$ ). Furthermore, Dunn's pairwise comparisons with Bonferroni adjustments show that the perceived driving performance ranking is higher for the virtual carpet concepts compared to the baseline concept (virtual carpet (basic):  $z = 2.9$ ,  $p = 0.013$ ; virtual carpet (longitudinal):  $z = 2.7$ ,  $p = 0.018$ ). Finally, Dunn's pairwise comparisons with Bonferroni adjustments also show that no significant difference in the ranking of perceived driving performance, perceived safety or perceived mental workload was found with respect to the comparison of the two virtual carpet concepts ( $z = 0.11$ ,  $p = 1.00$ ;  $z = 0.78$ ,  $p = 1.00$ ;  $z = 0.44$ ,  $p = 1.00$ ).



*Figure 31.* Means and standard errors (error bars) of the perceived driving performance ranking, the perceived safety ranking, and the perceived mental workload ranking for all AR HUD concepts. Significant interactions are shown when  $p < 0.05$ , when  $p < 0.01$ , and when  $p < 0.001$ .

## 5. DISCUSSION

The aim of this study was to address the gap in literature on the comparison of a virtual lead-vehicle AR HUD concept with other world-fixed AR HUD concepts. To do so we investigated the effects of different AR HUD concepts on lane position variability and mental workload. The AR HUD concepts that were presented were a baseline concept that only presented screen-fixed speed and navigation information, two virtual carpet concepts (one with longitudinal information and one without), and a virtual lead-vehicle concept. The virtual carpet concepts and the virtual lead-vehicle concepts all were presented together with the screen-fixed speed and navigation information. The current chapter will review the results, with the first section discussing the difference between the virtual lead-vehicle and virtual carpet (basic). Here, both the difference in lane position variability and mental workload are discussed between the virtual lead-vehicle and the virtual carpet (basic), next to the effects of these AR HUD concepts on the other exploratory variables. The second section discusses the effect of adding longitudinal color information to the virtual carpet concept on mental workload, lane position variability and the other exploratory variables. The third section compares the screen-fixed only concept with the other three concepts, investigating whether the world-fixed concepts are a benefit over screen-fixed only concept. Furthermore, the fourth section discusses some suggestions for further research on the AR HUD concepts and virtual environment that were not yet discussed in the first three sections. Finally, the fifth section discusses some general aspects that could be improved in future AR HUD concepts studies as well.

### 5.1 Virtual Lead-vehicle vs. Virtual Carpet (Basic)

#### *Mental Workload*

Following a lead-vehicle was expected to be more common to people compared to following a carpet, which was hypothesized to require less cognitive effort, and therefore lead to a decreased mental workload compared to the virtual carpet (basic) (H2). Another reason for a decrease in mental workload was since the virtual lead-vehicle provides longitudinal feedback to the driver, which makes it no longer needed for the driver to calculate the difference in speed limit by him- or herself. However, our data revealed that mental workload is significantly higher for the virtual lead-vehicle compared to the other three AR HUD concepts with the biggest difference in mental workload with the virtual carpet (basic). This effect is not only presented by the overall mental workload score, but also by its individual items focusing on the attentional mental workload, stress demand, temporal demand and the interference on the driving task caused by the navigation system. In addition, looking at the ranking data, perceived mental workload was ranked highest for the virtual lead-vehicle compared to the other three AR HUD concepts, and significantly higher compared to the virtual carpet concepts.

These findings suggest a higher mental workload for the virtual lead-vehicle not only compared to the virtual carpet (basic) but compared to all three other AR HUD concepts. This is different from the mental workload results of other virtual lead-vehicle studies, as both Topliss et al. (2019) and Pampel et al. (2019) showed improved or at least equal mental workload of the virtual lead-vehicle compared to other AR HUD concepts. However, these studies did not compare the virtual lead-vehicle with a virtual carpet or a similar screen-fixed concept as our baseline concept, making it difficult to compare these results. Namely, Topliss et al. (2019) compared a dynamic virtual lead-vehicle with a static virtual lead-vehicle and a static world-fixed arrow. Similarly, Pampel et al. (2019) compared the dynamic virtual lead-vehicle with a static arrow that fills up as the driver approaches the required turn. Therefore, our results show that although the virtual lead-vehicle scored lower in mental workload compared to screen-fixed arrow concepts and a static virtual lead-vehicle, there exist concepts that result in an even lower mental workload compared to the virtual lead-vehicle, such as the virtual carpet or our screen-fixed concept. However, to be able to make this comparison it is assumed that the virtual lead-vehicle in all virtual lead-vehicle studies including this study is similar, which is not true. Namely, the virtual lead-vehicle in the study by Pampel et al. (2019) stopped at the correct exit and was always 50 ft in front of the driven vehicle, where the virtual lead-vehicle in our study continued driving at after a maneuver and did not have a fixed distance from the driver. The virtual lead-vehicle's behavior in the

by Topliss et al. (2019) was explained to be similar to real-world vehicle's behavior. Our virtual lead-vehicle concept seems to be similar to theirs, however, it is not clear whether their vehicle drove through other vehicles and ignored traffic lights. In addition, our virtual lead-vehicle drove through the turns very quickly, where the virtual lead-vehicle from Topliss et al. (2019) probably used a more appropriate speed for in the turns.

It could be that some behaviors of the virtual lead-vehicle caused participants to pay more attention to the virtual lead-vehicle compared to the attention paid to the virtual carpet (basic) explaining the higher mental workload for the virtual lead-vehicle. First of all, from the debriefing it was clear that participants found it rather strange that the virtual lead-vehicle's speed in the turns was similar to its speed before approaching that turn. Although we aimed to create a behavior of the virtual lead-vehicle that resembled a real-world vehicle as much as possible, to create such behavior is complicated and time-consuming. In addition to this, it was not expected that this higher speed in the turns would have been as inconvenient as was found in the debriefing. However, it might be that mental workload increased because of this behavior, which is why it would be valuable to investigate a virtual lead-vehicle that drives at a more appropriate speed in the turns.

Second, participants indicated that the virtual lead-vehicle's braking lights were confusing as sometimes they were constantly brightened, but they did not fully understand why. Some participants thought that the brightened braking lights indicated that the participant should slow down, which resulted in a continuation of brightened braking lights, but in fact it occurred when the participant was driving too slow and therefore too far away from the virtual lead-vehicle. It would be interesting to investigate whether braking lights are required at all for a virtual lead-vehicle concept, since they might increase mental workload, while they might not improve driving performance or provide better navigation indications.

Third, the virtual lead-vehicle often accelerated slower than the driver, causing the driver to catch up with the virtual lead-vehicle or even drive through it. It also sometimes seemed to slow the driver down, such that he or she was able to keep an appropriate distance to the virtual lead-vehicle. During the pilot test and the creation of the experiment, this behavior was not experienced, therefore not corrected for before starting the experiment. It can be expected that this behavior has been very inconvenient and caused an increase in mental workload. As a solution, it might be better if the virtual lead-vehicle would drive at a fixed distance from the driver, such as the case in the study by Pampel et al. (2019). Also, it might be better to let the transparency of the virtual lead-vehicle become higher as you drive closer to the virtual lead-vehicle, or that it would teleport to a distance further away from the driver. However, such additional features might also increase mental workload, which is an expectation that should be investigated in future studies.

Fourth, the virtual lead-vehicle occluded more real-world information compared to the virtual carpet, especially when the driver was driving close to the virtual lead-vehicle. A solution for this would be to only present the virtual lead-vehicle closely before an upcoming maneuver or to increase its transparency. This would, however, result in a loss of its ability to provide longitudinal feedback. Another solution would be to adjust the transparency level of the vehicle such that it becomes more transparent. However, this should not decrease the visibility of the lead-vehicle to a level where it becomes hard to follow the navigation instructions of the lead-vehicle.

Fifth, the virtual lead-vehicle was designed to drive through other vehicles and to ignore the traffic lights. This behavior is different from real-world vehicles, which was indicated in the debriefing to be uncomfortable for some of the participants, as they were afraid that they would collide into the other traffic or ignore a traffic light. This might be a reason why the virtual lead-vehicle attracted more attention and therefore led to higher mental workload compared to the virtual carpet that. A solution to this might be to create a virtual lead-vehicle that resembles a real-world vehicle less, such that it drives at a fixed distance from the participant, however, then it would lose its ability to provide longitudinal feedback.

Finally, the lower mental workload of the virtual carpet (basic) could be that although it was initially assumed that a virtual lead-vehicle is more common to people, that in fact a virtual carpet is also quite common or even more common to people, since many people nowadays use a navigation system which includes a route line. Therefore, the virtual carpet concept might not have required more mental workload to understand how to follow it.

#### *Lane Position Variability*

It was also expected that the virtual lead-vehicle would lead to better driving performance, and therefore lower lane position variability, compared to the virtual carpet (basic) (H1). This was expected since the virtual lead-vehicle was assumed to be less abstract to interpret compared to the virtual carpet, as following a car also exists in real life. Because of this, drivers following the virtual lead-vehicle would be able to pay more attention to the driving task, causing less lane position variability compared to the virtual carpet (basic). However, our results show no significant difference in lane position variability between the virtual carpet concepts and the virtual lead-vehicle, but there was a trend in the opposite direction. The slightly better, but not significantly better, performance of the virtual carpet concepts compared to the virtual lead-vehicle could be because the carpets are visualized around the center of the lane, making it easier to use as a reference. The virtual lead-vehicle was also driving along the center of the lane; however, the virtual lead-vehicle might have been less easy to use as a reference since it indicates the center of the road less clearly compared to the virtual carpet. It is, again, difficult to compare these results with other studies. The study by Topliss et al. (2019) also did not show significant differences in lane position variability, however, their study did not involve a virtual carpet as comparison but only compared the virtual lead-vehicle with a static virtual lead-vehicle and a static world-fixed arrow. The studies by Pärsh et al. (2019) and Pfannmüller et al. (2015) did involve a virtual carpet similar to our virtual carpet (basic) concept but did not measure lane position variability.

#### *Lane Position*

With respect to the driving performance measures, the virtual lead-vehicle resulted in a 2.0 cm larger distance from the center of the lane compared to the virtual carpet (basic), however, similar to the difference in lane position variability, this difference was not significant. Similar to the explanation of the difference in lane position variability, this difference probably exists because the carpet can be used as a reference for driving close to the center of the lane.

#### *Speed*

No significant difference was found in average speed between the AR HUD concepts, which is similar as was found by Topliss et al. (2019). However, the virtual lead-vehicle did lead to a significantly lower speed limit exceedance (-8.4%) on the road fragments compared to the virtual carpet (basic). This could be caused by the fact that speeding while following the virtual lead-vehicle would result in the participant driving closer to the virtual lead-vehicle or even driving through the virtual lead-vehicle. It can be imagined that getting close to the virtual lead-vehicle is not desirable for drivers, as drivers are focused on maintaining distance between their own vehicle and the lead vehicle, causing them to slow down.

#### *Collisions*

Furthermore, a non-significant difference was present for the number of collisions with other vehicles. Namely, the virtual carpet (basic) induced three collisions, but the virtual lead-vehicle induced only one. It is not surprising that not a lot of collisions occurred, since this was the case for other studies as well (Kun, 2018; Medenica et al., 2011). However, the difference in collisions is still interesting, since it could potentially indicate that following a virtual lead-vehicle which drives through other traffic and traffic lights, which theoretically increases the chance of collisions, does not result in an actual increase in collisions compared to following a virtual carpet.

#### *Navigational Success*

Navigational success was similar for the virtual carpet (basic) and the virtual lead-vehicle, showing that both concepts function similarly with respect to navigation clarity. This is different from the study of Pampel et al. (2019) who found that following a virtual lead-vehicle led to higher navigation success compared to a traditional navigation system and an AR HUD concept with a static arrow that fills up as the driver approaches the required turn. Topliss et al. (2019) also measured navigation performance and found better performance for the static arrow compared to the dynamic virtual lead-vehicle or the static virtual lead-vehicle, which is similar to our study as it shows that the dynamic virtual lead-vehicles does not always induce the least navigation errors. Furthermore, Pärsch et al. (2019) found that their concept which was similar to our virtual carpet (basic) (Figure 4.b) led to barely any navigation errors (only one error), compared to other AR HUD concepts that induced five errors, which is similar to our results. Altogether, although the virtual lead-vehicle induced less navigation errors compared to other AR HUD concepts in the study of Pampel et al. (2019) and compared to the static virtual lead-vehicle in the study of Topliss et al. (2019), our findings show that both the virtual lead-vehicle and the virtual carpet (basic) score similarly good with respect to navigational success, with the virtual carpet (basic) possibly even better.

### *Perceived Safety*

Looking at the subjective variables, some additional results are worth mentioning. First of all, perceived safety is lower for the virtual lead-vehicle compared to the virtual carpet (basic) looking at both the perceived safety questionnaire and the perceived safety ranking. This could be explained by participants being aware that following the virtual lead-vehicle required a higher mental workload. Additionally, it might have felt unsafe for drivers to follow a virtual lead-vehicle at the moments where the lead-vehicle drove through other traffic or traffic lights. Next to that, the virtual lead-vehicle was the only world-fixed AR HUD concept that was presented as an overlay over the other vehicles, occluding some relevant road information. Looking at the virtual carpet (basic), the behavior was probably considered to be safer since the carpet did not involve complicated behavior or surprising events. Unfortunately, no similar studies have incorporated the measure of perceived safety, which is why these findings cannot be compared.

### *User Experience*

Considering user experience, the virtual lead-vehicle was found to have an excellent hedonic user experience, i.e., the originality of the design and aesthetics of the concept were very high. But, on the contrary, the virtual lead-vehicle's pragmatic user experience was benchmarked as bad. These results are opposite for the virtual carpet (basic), for which the hedonic user experience is below average and significantly lower than the virtual lead-vehicle's hedonic user experience. Its pragmatic user experience is benchmarked as excellent and significantly better than the virtual lead-vehicle's pragmatic user experience. Overall, combining the hedonic and pragmatic user experience, the virtual lead-vehicle was evaluated as providing a bad user experience, and the virtual carpet (basic) as good user experience. One could expect that the virtual lead-vehicle's bad pragmatic user experience is possibly caused by some of its unexpected behaviors. Next to that, it could be because the virtual lead-vehicle ignored other traffic and traffic lights. However, since no qualitative data is collected in this study, we cannot be sure whether these assumptions are correct. Other studies on the virtual lead-vehicle did not measure user experience. However, Pampel et al. (2019) did perform a preference ranking between a virtual lead-vehicle, a traditional navigation system, and a static arrow that fills up as the driver approaches the required turn. Their preference ranking, however, did not show a difference in preference between the navigation systems.

### *Ranking of Perceived Driving Performance*

Finally, participants ranked their driving performance to be worst when using the virtual lead-vehicle as an AR HUD concept, followed by the baseline concept and the two virtual carpet concepts. This is an interesting result, since the virtual lead-vehicle never scores worst for the objective driving performance measures. It seems, therefore, that the bad usability and perceived safety impact the overall subjective evaluation of driving performance. No other virtual lead-vehicle studies evaluated the ranking of perceived driving performance.

Altogether, it can be concluded that although no clear differences in driving performance were found between the virtual carpet (basic) and the virtual lead-vehicle, the subjective results clearly show a preference towards the virtual carpet (basic). More specifically, where the hedonic user experience score indicates that the virtual lead-vehicle might be very attractive for people at first, its bad user experience, low perceived safety and higher mental workload compared to the virtual carpet (basic) suggest that the virtual carpet (basic) might be a better world-fixed AR HUD concept. These findings seem to resemble the destiny of the AR Fox, Google's pedestrian guide, which was enthusiastically received at first, but in the end turned out to be too distracting (Stein, 2019). The virtual lead-vehicle's only advantage over the virtual carpet is that it induces less speed limit exceedance. However, this benefit does not outweigh the negative effects of the virtual lead-vehicle on mental workload and the other subjective measures.

## 5.2 Longitudinal Information

Since the virtual lead-vehicle provides longitudinal feedback, it was interesting to investigate the effect of adding longitudinal information to world-fixed AR HUD concepts other than the virtual lead-vehicle. Therefore, the difference between the virtual carpet (basic) and virtual carpet (longitudinal) was investigated as well. In addition, this section also covers the difference between the virtual carpet (longitudinal) and the virtual lead-vehicle. Where possible, the results are compared to previous studies, however, only one study was found that has evaluated the addition of longitudinal information.

### *Mental workload*

Similar to the virtual lead-vehicle, it was assumed that adding longitudinal information, i.e., changing the virtual carpet's color to yellow, orange and ultimately red when the driver exceeds the speed limit, results in a lower mental workload (H4). This was expected as the driver no longer has to calculate the difference between the driven speed and the speed limit by him/herself. However, our data seem to suggest an opposite effect. Namely, the overall mental workload is significantly higher when adding longitudinal information to the virtual carpet compared to when not. This difference is present for each individual mental workload item except for the auditory mental workload but is only significant for the stress demand. However, these results should be taken with care, as the ranking of mental workload does not show a significant difference between the virtual carpet (basic) and the virtual carpet (longitudinal). Altogether, our data shows contradictory results with respect to the effect of adding longitudinal information on mental workload. No other studies were found that investigated the effect of longitudinal information on mental workload, making it impossible to compare our results with other studies.

If adding longitudinal information to the virtual carpet indeed causes increased mental workload, we can think of multiple explanations. First of all, it can be quite mentally demanding if you are constantly being reminded about the speed limit, instead of only when you feel the desire to check the speed limit.

In addition, from debriefing it was understood that for some participants the color changing was a bit too strict. The color of the carpet was set such that it was blue when below the speed limit; yellow when it was 3% above the speed limit; orange when it was 6% above the speed limit, and finally it turned red when it was 9% above the speed limit. However, although the yellow color was just meant to be a warning, some participants might have considered it annoying to have a warning after 3% of speed limit exceedance and might prefer it when the color would only change at a higher threshold. It could be that when experiencing the colors as too strict, that it also drew more attention, and therefore resulted in higher mental workload compared to the virtual carpet without longitudinal information. However, there were also participants indicating that they did not find the color changing too strict. Therefore, we suggest that, when longitudinal information would be implemented in a virtual carpet, drivers should be able to set the strictness of the coloring by themselves.

Furthermore, some participants indicated that they wondered during the drive whether the virtual carpet would also provide longitudinal feedback when approaching a turn, which is often seen in video racing games. This was, however, similar as with the virtual lead-vehicle, not implemented. Possibly, this indicates that participants have been figuring out the functionality of the virtual carpet (longitudinal) longer than the virtual carpet (basic) longitudinal information, which might explain the higher mental workload of the virtual carpet (longitudinal). It would, therefore, be interesting to perform a study in which participants drive multiple times using more complicated AR HUD concepts as the virtual carpet (longitudinal) to make sure mental workload is not higher because of this learning effect.

Finally, the longitudinal information added to the virtual carpet was not always accurate. In particular, there was a delay of the color changing back to blue after the speed limit was no longer exceeded. In rare occasions, the carpet turned red when the participant slightly touched a road part where the speed limit was lower than the speed limit on the road he or she was driving on. During the development of the concept and during the pilot test this ambiguous behavior of the carpet was not noticed, therefore not altered. However, it might have confused the participant about the functionality of the carpet, therefore increasing the mental workload compared to the virtual carpet without longitudinal information.

#### *Lane Position Variability*

Similar to our reasoning for a decreased mental workload, we expected a decrease in lane position variability when adding longitudinal color information to the virtual carpet concept (H3). This was expected since the driver did not have to calculate speed limit exceedance by himself/herself, making more cognitive capacity free for focusing on the primary driving task. However, lane position variability was found to be not significantly different between the two virtual carpet concepts. The results of lane position variability are also different from Hagen et al (2005), who found that when presenting speed information on a HUD, participants were better at monitoring their speed, but worse in maintaining their lane position. They explained this decrease in maintaining lane position due to cognitive tunneling to the HUD. Possibly, since our longitudinal concept included a virtual carpet and this was not present in the concept of Hagen et al. (2005), it shows that the carpet functioned as a reference to maintain a stable lane position very strongly. Furthermore, although mental workload is higher when adding longitudinal information to the virtual carpet concept, it seems to not result in a mental overload, since the driver was able to maintain a stable lane position.

#### *Driving Performance and Navigation Performance*

Considering the exploratory driving performance variables and navigation performance, no significant differences are found between the two virtual carpet AR HUD concepts. More specifically, no significant difference was found for average lane position and average speed. Although speed limit exceedance was 5.2% lower for the virtual carpet with longitudinal color information, this difference was not statistically significant. If this difference would have been significant, it would have been similar to the finding by Hagen et al. (2005) who found that when presenting speed information on a HUD, participants were better at monitoring their speed, i.e., they showed less deviation from the speed limit. Furthermore, no significant difference in navigation errors between the two virtual carpet concepts was present. The virtual carpet (basic) concept did induce three collisions, where the virtual carpet (longitudinal) did not induce any collision at all. However, no conclusions can be made from this as this was not significantly different.

#### *User Experience, Perceived Safety and Perceived Driving Performance*

Looking to the subjective data, it was found that, similar to the virtual lead-vehicle, the hedonic user experience of the virtual carpet (longitudinal) was significantly higher compared to the virtual carpet (basic). In addition, the hedonic user experience of the virtual carpet with longitudinal information was only slightly, and not significantly lower than for the virtual lead-vehicle. However, when considering the pragmatic and overall user experience, the virtual carpet (longitudinal) has a significantly better user experience compared to the virtual lead-vehicle and similar to the virtual carpet (basic). The absence of difference in overall user experience between the two virtual carpet concepts is interesting as it therefore seems that the earlier mentioned higher mental workload for the longitudinal

information does not seem to induce a worse user experience. In addition, it can be seen from the results that the two virtual carpet AR HUD concepts do not show any other significant difference with respect to the perceived safety and the rankings of perceived driving performance and perceived safety.

Altogether, our results suggest that following the virtual carpet with longitudinal information might have been more mentally demanding compared to following the virtual carpet without this information, but that it did not result in a mental overload as driving performance was similar for both virtual carpet concepts. In addition, subjective measures such as pragmatic user experience and perceived safety also did not differ between the two virtual carpet concepts. Furthermore, the pragmatic user experience is higher for the virtual carpet with longitudinal information, which might indicate that participants would be eager to try the extra longitudinal functionality. Also, since speed limit exceedance was decreased (not significantly) with 5.2%, we conclude that adding longitudinal information might be a good addition to the virtual carpet, but it should be kept in mind that it might increase mental workload. However, when it is aimed to keep the mental workload as low as possible, it might be better to not include the longitudinal information. In addition, the similarity of results between the two virtual carpet concepts shows that the negative results of the virtual lead-vehicle compared to the virtual carpet (basic) concept are probably not caused by the virtual lead-vehicle's longitudinal feedback (i.e., the vehicle coming closer to the driver or further away is an indication of speeding or driving to slow). Only the higher mental workload of the virtual lead-vehicle can possibly be explained by the virtual lead-vehicle's longitudinal feedback, since this is higher for both the virtual carpet (longitudinal) and virtual lead-vehicle, but as mentioned above there also are other possible explanations for the higher mental workload of the virtual lead-vehicle.

### 5.3 Screen-fixed With or Without World-fixed

Several results are interesting with respect to the difference in presenting a concept with only a screen-fixed HUD with speed and navigation information, or when providing both screen-fixed information together with world-fixed navigation like the virtual carpet and the virtual lead-vehicle. This is interesting, since studies have shown mixed results with respect to the advantage of world-fixed concepts over screen-fixed concepts (Merenda et al., 2018; Pärsch et al., 2019; Pfanmüller et al., 2015). However, no study seems to have examined the effect of both screen-fixed and world-fixed AR HUD concepts.

Although a screen-fixed AR HUD concept is not present in everyone's car already, the baseline AR HUD concept shows a bad hedonic user experience, meaning that participants did not find the concept innovative or aesthetically pleasing. This can possibly be due to an anchoring effect, as all four concepts had been explained before the experiment started, making it possible that the three world-fixed AR HUD concepts were functioning as an anchor, with which the screen-fixed HUD was being compared. We expect that this might have resulted in a lower hedonic user experience score than if the baseline concept would have been compared with for example a more traditional navigation system such as a navigation system located at the center of the dashboard. Similar results were, however, found in a study by Pfanmüller et al. (2019), in which lower attractiveness was found for a screen-fixed arrow and navigation information concept compared to a concept similar to the virtual carpet concept in this study.

In addition, this study reveals that the screen-fixed only concept leads to a higher lane position variability compared to the screen-fixed concepts that are presented together with world-fixed concepts. This difference is, however, only significant when comparing with the virtual carpet concept. The higher lane position variability for the baseline concept could be because the **cognitive distance** from the navigation information to the road is larger compared to the world-fixed concepts where the navigation information is projected onto the real-world. In these world-fixed concepts, screen-fixed information was also presented, however, probably the screen-fixed navigation and speed information was used less, since significantly less navigation errors have been made using the world-fixed concepts compared to the screen-fixed only concept. Namely, 13 navigation errors were made for the baseline

concept and only 2 for all three world-fixed concepts together. However, that participants used the screen-fixed informed less when presented next to world-fixed information might also be because participants had difficulty in understanding the screen-fixed HUD's distance-to-turn counter, which was understood from the debriefing. In hindsight, it appeared that the distance-to-turn counter was slightly later than participants expected it to be, which is why participants sometimes realized too late that they had arrived at the location where a maneuver was required. In practice, participants quickly understood the delay of these maneuvers, which is why most of the mistakes only occurred in the first turn of the route.

For the other variables, however, the baseline concept does not perform clearly better or worse compared to the other three AR HUD concepts. For example, the baseline concept has 1.7% lower speed limit exceedance compared to the virtual carpet (basic), but 3.5% and 5.4% more speed limit exceedance compared to the virtual carpet (longitudinal) and virtual lead-vehicle. This difference is, however, not significant. Similarly, the baseline concept's pragmatic user experience is lower compared to the world-fixed virtual carpet concepts, but significantly better compared to the world-fixed virtual lead-vehicle concept. Similar results were found in a study by Pfanmüller et al. (2019), who found a higher perceived functionality of a virtual carpet concept compared to a screen-fixed arrow and navigation information concept. Furthermore, no clear difference was found when adding world-fixed navigation information with respect to perceived safety. This finding is similar for some of the mental workload items, as mental workload was found to be significantly higher for the baseline concept compared to the virtual carpet (basic), but significantly lower compared to the virtual lead-vehicle.

Altogether, since both world-fixed concepts show better navigation performance and lane position variability compared to the screen-fixed-only concept, and the virtual carpet concept shows also better performance for mental workload, perceived safety and user experience, we can conclude that presenting a combination of world-fixed and screen-fixed AR HUD navigation information is better compared to presenting only screen-fixed information. Here, we do however note, that the virtual lead-vehicle might not be the best world-fixed concept to add to the screen-fixed navigation information.

## **5.4 Further Reflection on AR HUD Concepts and Virtual Environment**

In addition to the above reflections done on the AR HUD concepts, we have some further suggestions for investigation of the AR HUD concepts. First of all, both the virtual lead-vehicle and virtual carpet are presented all the time. However, when, for example, the driver is driving on a straight road such as a highway, it might not be necessary to have a virtual lead-vehicle presented in front of the car. Therefore, it could be argued that the world-fixed navigation visualizations would actually only be required in case a maneuver has to be performed soon. The speed limit exceedance would not be at risk, since drivers could use their cruise control on these long and straight roads and thus are not in need of the longitudinal information. Additionally, screen-fixed speed and navigation information could still be presented, therefore it can still be found when a maneuver is about to happen. Future research should investigate whether indeed world-fixed AR HUD navigation should only be projected when a maneuver is about to happen.

In addition, next to investigating the transparency level of the virtual lead-vehicle, this is also required for the virtual carpet. Since it might be necessary to use a carpet with higher transparency than was used in this study, since otherwise road markings might be too difficult to notice. However, as no road markings except give-way road-markings were present in our environment, this has probably had no effect on our results. Therefore, a future study is desired which would investigate whether the results on measures such as user experience and driving performance would improve when transparency of the virtual carpet is decreased.

Concerning the virtual environment, this study improved previous virtual lead-vehicle driving simulator studies by adding other vehicles in the environment. We believe that this was indeed a good addition to our method, since it resulted in a more realistic scenario. Also, it gave us the opportunity to

investigate whether drivers would collide into other traffic while using the different AR HUD navigation concepts. One disadvantage of this is that the addition of traffic makes it more complicated to compare our results with previous studies on the virtual lead-vehicle, but we believe that the benefit of increased realism outweighs this.

Furthermore, the video-based study by Topliss et al. (2019) showed that mainly for complex intersections, following the instructions of the virtual lead-vehicle was beneficial, however, we have not been able to investigate this further. Because of time limitations, it was decided to use the Dutch city scene 'CityPlusHighway' from Green Dino B.V. An elaborate description of the scene can be found in Section 3.4.1. Although the environment was pretty detailed and well-developed, it was difficult to implement changes to the environment. For a future study on the virtual lead-vehicle, we would advise to use an environment with more complex intersections, multi-lane roads and ambiguous turns, such that the findings of Topliss et al. (2019) can be explored further.

## 5.5 Suggestions for Future Research

Finally, we want to present several limitations of our method, which could be improved in future studies investigating AR HUD concepts or studies that specifically want to improve this study. First of all, the AR HUD concept was intended to be a replacement for any other navigation system present in the car. However, nothing regarding the presence of additional navigation systems was explained to participants. This makes it unclear what concept people had in mind when answering questions about the navigation system. In hindsight, it would have been good to explain the idea of using an AR HUD concept more clearly. For example, it might be that participants missed knowledge about future turns, similar as in the study by Topliss et al. (2019). However, participants that expected that a dashboard display would always be present as well when this concept would be implemented in a real car, might not have considered this a problem of the AR HUD concept. Next to clarifying the use of the AR HUD concepts in future studies, research could also be useful in which the focus lies on how to use the AR HUD concept could also be useful: together with a dashboard navigation system, or without? In addition, it should be investigated on what display which information should be presented.

We used virtual AR HUD concepts in a driving simulator instead of real AR HUD technology. The use of virtual AR HUD concepts was valuable since it made it possible to compare AR HUD concepts that were not limited by technical capabilities such as the limited field of view. However, depending on the future of AR HUD technology, i.e., whether AR HUD concepts continue to be limited by the field of view, it might also be interesting to investigate the AR HUD concepts in our study using real AR HUD technology. In addition, it might be that in practice it will not be possible to present a virtual carpet underneath other vehicles similar as in both our AR HUD concepts. The study by Pärssch et al. (2015) found that when a carpet covered the other vehicles, it led to more navigation errors and was more disturbing compared to below the other vehicles. However, it did not result in a difference in brake responses. This suggests that in case future AR HUD technology only allow the virtual carpet to be presented above other vehicles, a future study should investigate the driving performance of participants using this virtual carpet further.

Furthermore, it would be good to gain insights into whether drivers following a virtual lead-vehicle, or a virtual carpet would notice certain unexpected events occurring on the road, which could be obtained by measuring situation awareness or eye gaze information. This is because mental workload has shown to mainly not affect lateral driving performance. However, it might be that the world-fixed AR HUD concepts do affect other primary driving tasks for which situation awareness is required, such as sudden road events.

Finally, this study used a quantitative approach for evaluating the different AR HUD concepts. Although quantitative measures can predict human behavior on the road more clearly compared to qualitative measures, without qualitative measures it is more difficult to interpret our findings. For example, user experience and perceived safety show very clear differences between the AR HUD

concepts but based on observations and debriefing we can only guess why this would be the case. Besides that, we have not been able to investigate participant's thoughts on the design of the screen-fixed navigation information. In a future study, it would be interesting to perform short interviews with the participants such that more knowledge is gained about specific behaviors of the AR HUD concepts.

## 6. CONCLUSIONS AND DESIGN RECOMMENDATIONS

This study contributes to the automotive industry and automotive research field by investigating the effects of different AR HUD concepts on lane position variability and mental workload, and several other exploratory variables. More specifically, it investigated the effectiveness of using a virtual lead-vehicle as a world-fixed AR HUD navigation concept. Different from other lead-vehicle studies, our study compared the virtual lead-vehicle with another world-fixed AR HUD concept: the virtual carpet. First of all, based on the high number of navigation errors, it is recommended to present the screen-fixed AR HUD navigation concept together with world-fixed AR HUD. In addition, although this study did not look at AR HUD concepts without screen-fixed navigation and speed information, we would recommend following the findings from previous studies and use screen-fixed information, such that drivers are aware of future maneuvers.

Furthermore, our results show that the virtual carpet is a more promising world-fixed AR HUD concept compared to the virtual lead-vehicle. This is because the virtual carpet concept (basic) resulted in lower mental workload, higher user experience, higher perceived safety and was ranked higher with respect to perceived driving performance compared to the virtual lead-vehicle. These differences, however, were probably not due to the addition of longitudinal feedback in the virtual lead-vehicle concept, since no differences between these variables were found between the virtual carpet concepts except for mental workload. Next, the virtual carpet (basic) not only showed to be better than the virtual lead-vehicle, but also compared to the screen-fixed only concept, since it improved user experience, perceived driving performance and navigation errors. However, it should be noted that the virtual carpet concept was presented underneath other vehicles, where this might not be possible using real AR HUD technology, suggesting that further research is recommended to figure out whether a virtual carpet presented on top of other vehicles would still be significantly better than the virtual lead-vehicle. We expect that the virtual carpet would still be better compared to the current virtual lead-vehicle concept, but possibly this difference would be smaller when the virtual lead-vehicle's behaviour is improved.

The virtual lead-vehicle did show proper navigation abilities, but more research is needed to improve the concept and investigate its effect on driving performance. First of all, the virtual lead-vehicle might still be advantageous at complex intersections such as was found by Topliss et al. (2019), but which was not covered in this study. Besides that, although we assume that the virtual lead-vehicle's indicator lights and color have been designed properly and should be kept in future designs, some of the virtual lead-vehicle's behaviour in this study could be improved and should be explored further. For example, the braking lights should flicker less, the virtual lead-vehicle should use an appropriate speed when driving through turns, and the virtual lead-vehicle should accelerate quicker. In addition, deliberate behaviors of the virtual lead-vehicle such as the low transparency level, its ignoring of other traffic and traffic lights, or that it drove according to the speed limit, should be investigated further. It could also be investigated whether the virtual lead-vehicle should have a fixed distance to the driver, such that it does not provide longitudinal information at all. Finally, it might be better to only present the virtual lead-vehicle at the moment of an upcoming maneuver, such that it is not distracting when driving on a straight road.

We assume our virtual carpet design to be recommendable for the automotive industry (i.e., blue, transparent and an in-game width of 140 cm). However, it is advisable to further investigate an appropriate transparency level, since the carpet should not occlude road marks. Besides that, although not investigated in our study, similar to the virtual lead-vehicle, we think that it might not be necessary to present the virtual carpet at all times, i.e., it might be better to only present the carpet close to an upcoming maneuver. The addition of longitudinal information to the virtual carpet resulted in an increase in mental workload, but probably not in an overload, since lateral driving performance was not significantly affected. However, as mental workload is increased, using a virtual carpet with longitudinal information is not yet recommended. However, if further research indicates that longitudinal information can be used safely, we advise to make the addition of longitudinal color information a setting that users could activate or deactivate, based on their preference. Additionally, the

strictness level (i.e., at what percentage of speed limit exceedance the carpet's color changes) could also be a feature that can be adjusted with regard to the user's preference. Future research can investigate further whether strictness and color addition might indeed be preferred to be a setting.

Altogether, we propose that the virtual carpet is the most promising and ready to use world-fixed AR HUD concept. We envision the virtual carpet to be presented together with screen-fixed speed and distance-to-turn information, but for a wholistic view of the different in-vehicle systems, research is needed to figure out what information should be presented on what display exactly. Next to that, the virtual carpet's effects on driving performance should be investigated when it is presented as an overlay over other vehicles, similar to the virtual lead-vehicle. Therefore, we still envision a future for the virtual lead-vehicle, but it probably requires more improvements and further investigation, since the current concept still includes several ambiguities.

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## APPENDICES

## Appendix A: DALI Questionnaire

**Please answer the following questions about the driving and navigation task.**

For each factor, please indicate the level of constraint (in Dutch: "beperking") felt during the test on a scale from 0 (very low level of constraint) to 5 (very high level of constraint), with regard to the driving and navigation task.

**1. Attentional demand:** please indicate the mental (i.e., to think about, to decide...), visual and auditory demand required during the driving task.

(0 = Low)                                              (5 = High)

0          1          2          3          4          5

**2. Visual demand:** please indicate the visual demand required during the driving task.

[illegible]

**3. Auditory demand:** please indicate the auditory demand required during the driving task.

(0 = Low)                      (5 = High)

0          1          2          3          4          5

**4. Stress demand:** please indicate your level of stress (i.e., fatigue, insecurity, irritation, feelings of discouragement) during the driving task.

(0 = Low) 0 1 2 3 4 (5 = High)

**5. Temporal demand:** Please indicate the constraints (in Dutch: "beperkingen") felt due to time pressure when following the navigation instructions during the driving task.

(0 = Low)                      (5 = High)

0          1          2          3          4          5

**6. Interference:** Please indicate the disturbance to the driving task caused by using the navigation system.

(0 = Low)                      (5 = High)

0          1          2          3          4          5

## Appendix B: Perceived Safety Questionnaire

**Please answer the following questions about the navigation system.**

1. I believe that using the navigation system might be dangerous.

(1 = Extremely disagree)      1      2      3      4      5      6      (7 = Extremely agree)  
7

2. Using the navigation system does not really bother me to drive.

(1 = Extremely disagree)      1      2      3      4      5      6      (7 = Extremely agree)  
7

3. I believe using this navigation system is safe.

(1 = Extremely disagree)      1      2      3      4      5      6      (7 = Extremely agree)  
7

## Appendix C: UEQ-S Questionnaire

### Please answer the following statements about the navigation system.

The next questionnaire consists of pairs of contrasting attributes that may apply to the navigation system. Please express your agreement with the attributes by ticking the circle that most closely reflects your impression.

1. The navigation system is...

(1 = Obstructive)                      (7 = Supportive)  
1            2            3            4            5            6            7

2. The navigation system is...

(1 = Complicated)                      (7 = Easy)  
1            2            3            4            5            6            7

3. The navigation system is...

(1 = Inefficient)                      (7 = Efficient)  
1            2            3            4            5            6            7

4. The navigation system is...

(1 = Confusing)                      (7 = Clear)  
1            2            3            4            5            6            7

5. The navigation system is...

(1 = Boring)                      (7 = Exciting)  
1            2            3            4            5            6            7

6. The navigation system is...

(1 = Not interesting)                      (7 = Interesting)  
1            2            3            4            5            6            7

7. The navigation system is...

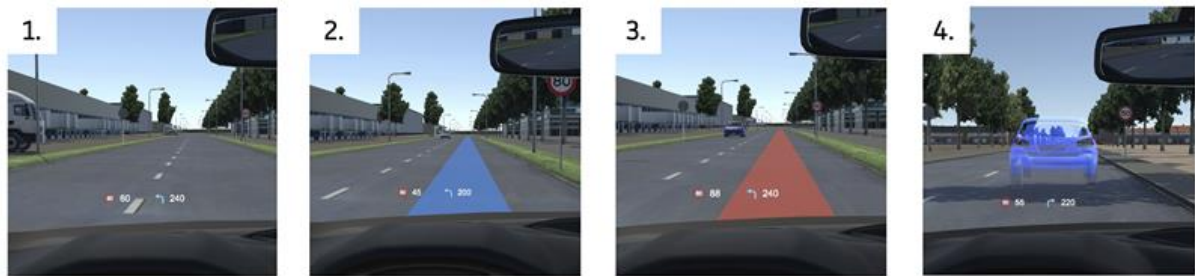
(1 = Conventional)                      (7 = Inventive)  
1            2            3            4            5            6            7

8. The navigation system is...

(1 = Usual)                      (7 = Leading edge)  
1            2            3            4            5            6            7

## Appendix D: Ranking Questionnaire

Please answer the following questions with regard to your preference for all conditions. Please use the pictures below to recall the conditions and use the numbering of the conditions in the image below.



### 1. Please rank the conditions based on your perceived driving performance using the navigation system. (I.e. controlling the car and following the driving rules)

*Top = condition with best driving performance, bottom = condition with worst driving performance*

Condition 1 - Speed and direction information

Condition 2 - Speed and direction information + Virtual carpet (always blue)

Condition 3 - Speed and direction information + Virtual carpet (changes colors)

Condition 4 - Speed and direction information + Virtual lead-vehicle

### 2. Please rank the conditions based on how safe it felt using the navigation system.

*Top = condition with highest perceived safety, bottom = condition with lowest perceived safety*

Condition 1 - Speed and direction information

Condition 2 - Speed and direction information + Virtual carpet (always blue)

Condition 3 - Speed and direction information + Virtual carpet (changes colors)

Condition 4 - Speed and direction information + Virtual lead-vehicle

### 3. Please rank the conditions based on the amount of mental workload needed to follow the navigation instructions. (I.e. how much mental attention or thinking you needed to do to follow the navigation)

*Top = condition with highest mental workload, bottom = condition with lowest mental workload*

Condition 1 - Speed and direction information

Condition 2 - Speed and direction information + Virtual carpet (always blue)

Condition 3 - Speed and direction information + Virtual carpet (changes colors)

Condition 4 - Speed and direction information + Virtual lead-vehicle

## Appendix E: C# Scripts

### ChangeColor.cs

```
1. using System.Collections;
2. using System.Collections.Generic;
3. using UnityEngine;
4.
5. public class ChangeColor : MonoBehaviour
6. {
7.     public int speedLimit;
8.     public float carSpeed;
9.     public Color currentColor;
10.    public Color targetColor;
11.
12.    public Color redColor;
13.    public Color orangeColor;
14.    public Color yellowColor;
15.    public Color blueColor;
16.
17.    public CarInfoManager carInfo;
18.    public EnvironmentInfoManager environmentInfo;
19.
20.    Renderer rend;
21.
22.    private void Start()
23.    {
24.        rend = GetComponent<Renderer>();
25.        carInfo = GameObject.Find("HUD").GetComponent<CarInfoManager>();
26.        environmentInfo =
27.        GameObject.Find("HUD").GetComponent<EnvironmentInfoManager>();
28.        rend.material.SetColor("_Color", new Color(255f / 255f, 0f / 255f, 0f / 255f,
29.        0.6f));
30.
31.        redColor = new Color(255f / 255f, 0f / 255f, 0f / 255f, 0.6f);
32.        orangeColor = new Color(255f / 255f, 140f / 255f, 0f / 255f, 0.6f);
33.        yellowColor = new Color(255f / 255f, 215f / 255f, 0f / 255f, 0.6f);
34.        blueColor = new Color(30f / 255f, 144f / 255f, 255f / 255f, 0.6f);
35.    }
36.
37.    private void Update()
38.    {
39.        carSpeed = carInfo.carSpeed;
40.        speedLimit = environmentInfo.speedLimit;
41.
42.        if (carSpeed > (speedLimit * 1.09))
43.        {
44.            targetColor = redColor;
45.            rend.material.SetColor("_Color", targetColor); //set color to red when
46.            speedLimit is exceeded with 109%.
47.        }
48.        else if (carSpeed > (speedLimit * 1.06))
49.        {
50.            targetColor = orangeColor;
51.            rend.material.SetColor("_Color", targetColor); //set color to orange when
52.            speedLimit is exceeded with 106%.
53.        }
54.        else if (carSpeed > (speedLimit * 1.03))
55.        {
56.            targetColor = yellowColor;
57.            rend.material.SetColor("_Color", targetColor); //set color to yellow when
58.            speedLimit is exceeded with 103%.
59.        }
60.        else
61.        {
62.            targetColor = blueColor;
63.            rend.material.SetColor("_Color", targetColor); //set color to blue when
64.            speed is good
65.        }
66.
67.        currentColor = Color.Lerp(currentColor, targetColor, 0.95f * Time.deltaTime);
68.        //change the color more gradually.
69.        rend.material.color = currentColor;
70.    }
71. }
```

```
64. }
```

## VCFollowRoute.cs

```
1. using System.Collections;
2. using System.Collections.Generic;
3. using UnityEngine;
4. using PathCreation;
5.
6. public class VCFollowRoute : MonoBehaviour
7. {
8.     public PathCreator pathCreator;
9.
10.    public float speedLimit; //speedlimit obtained at driver position
11.    public float targetSpeed;
12.    public float currentSpeed;
13.
14.    public Vector3 driverLocation;
15.    public Vector3 virtualCarLocation;
16.    [SerializeField] private float maximumDistance;
17.
18.    float distanceTravelled; //from the original pathCreator code
19.    public CarInfoManager carInfo; //needed to get driver location
20.    public EnvironmentInfoManager environmentInfo; //needed to obtain the speedlimit
21.
22.    private void Start()
23.    {
24.        carInfo = GameObject.Find("HUD").GetComponent<CarInfoManager>(); //needed to
        get driver location
25.        environmentInfo =
        GameObject.Find("HUD").GetComponent<EnvironmentInfoManager>(); //obtain speedlimit
26.        targetSpeed = 10;
27.    }
28.
29.
30.    void Update()
31.    {
32.
33.        maximumDistance = speedLimit * 0.56f; //sets the maximum distance between the
        lead-vehicle and the driver to speedLimit * 0.56
34.        driverLocation = carInfo.carPos; //to get position of driver.
35.        virtualCarLocation = transform.position; //to get position of virtual lead-
        vehicle
36.
37.
38.        //When the distance between the virtual lead-vehicle and the driver is larger
        than the maximum distance, then the virtual car brakes.
39.        //Otherwise, the virtual car will drive at a speed that is 1.05 * speedLimit.
40.        if (Vector3.Distance(driverLocation, virtualCarLocation) > maximumDistance)
41.        {
42.            targetSpeed = 0;
43.
44.        }
45.        else if (Vector3.Distance(driverLocation, virtualCarLocation) <
        maximumDistance)
46.        {
47.            targetSpeed = ((speedLimit / 3.6f) * 1.05f);
48.        }
49.
50.        currentSpeed = Mathf.Lerp(currentSpeed, targetSpeed, 0.5f * Time.deltaTime);
        //gradually changes speed of virtual car
51.
52.        //the following lines of code are from the original pathCreator code, adjusted
        such that the virtual car follows the route, created with the pathCreator tool.
53.        distanceTravelled += currentSpeed * Time.deltaTime;
54.        transform.position = pathCreator.path.GetPointAtDistance(distanceTravelled);
55.        transform.rotation =
        pathCreator.path.GetRotationAtDistance(distanceTravelled);
56.
57.    }
```

```
58. }
```

## VCLeftLight.cs

```
1. using System.Collections;
2. using System.Collections.Generic;
3. using UnityEngine;
4.
5. public class VCLeftLight : MonoBehaviour
6. {
7.     public string manœuvre;
8.     public float Timer;
9.     public float interval = 0.5f;
10.
11.     public float thresholdShowLeftCarLight = 200f;
12.     public float distance;
13.
14.     private Behaviour halo;
15.
16.     public RoutingEngine routingEngine;
17.
18.     void Start()
19.     {
20.         routingEngine = GameObject.Find("Condition3").GetComponent<RoutingEngine>();
21.         halo = (Behaviour)GetComponent("Halo");
22.     }
23.
24.     private void Update()
25.     {
26.         // retrieve distance from car to next instruction
27.         distance = routingEngine.distance;
28.
29.         // if distance from car to instruction is less than threshold, go to next
instruction in the list
30.         if (distance < thresholdShowLeftCarLight)
31.         {
32.             EnableLights();
33.         }
34.         else
35.         {
36.             halo.enabled = false;
37.         }
38.     }
39.
40.     void EnableLights()
41.     {
42.         manœuvre = routingEngine.manœuvre;
43.
44.         if (manœuvre == "LEFT")
45.         {
46.             FlickerLight();
47.         }
48.         else if (manœuvre == "EXIT_LEFT")
49.         {
50.             FlickerLight();
51.         }
52.         else if (manœuvre == "FINISH_LEFT")
53.         {
54.             FlickerLight();
55.         }
56.         else if (manœuvre == "ROUNDABOUT_270")
57.         {
58.             FlickerLight();
59.         }
60.         else
61.         {
62.             halo.enabled = false;
63.         }
64.     }
65.
66.     void FlickerLight()
```

```

67.     {
68.         if (Timer > 0)
69.             Timer -= Time.deltaTime;
70.         if (Timer <= 0)
71.         {
72.             halo.enabled = !halo.enabled;
73.             Timer = interval;
74.         }
75.     }
76. }
77. }

```

## VCbrakelight.cs

```

1. using System.Collections;
2. using System.Collections.Generic;
3. using UnityEngine;
4.
5. public class VCbrakelights : MonoBehaviour
6. {
7.     public VCFollowRoute vcFollowRoute;
8.     public float acceleration;
9.     public float lastVelocity;
10.    private Behaviour halo;
11.
12.    private void Start()
13.    {
14.        vcFollowRoute = GameObject.Find("VirtualCar").GetComponent<VCFollowRoute>();
15.        halo = (Behaviour)GetComponent("Halo");
16.    }
17.
18.    private void Update()
19.    {
20.        acceleration = (vcFollowRoute.currentSpeed - lastVelocity) /
Time.fixedDeltaTime;
21.        lastVelocity = vcFollowRoute.currentSpeed;
22.
23.        if (acceleration > 0)
24.        {
25.            halo.enabled = false;
26.        }
27.        else
28.        {
29.            halo.enabled = true;
30.        }
31.    }
32. }

```

## Appendix F: Summary Statistics

*Means (M) and standard deviations (SD) of all variables per AR HUD concept*

| Variable                      | Baseline |      |      | Virtual carpet (basic) |      |      | Virtual carpet (longitudinal) |      |      | Virtual lead-vehicle |      |      |
|-------------------------------|----------|------|------|------------------------|------|------|-------------------------------|------|------|----------------------|------|------|
|                               | n        | M    | SD   | n                      | M    | SD   | n                             | M    | SD   | n                    | M    | SD   |
| Lane position variability (m) | 33       | 0.29 | 0.08 | 33                     | 0.29 | 0.09 | 32                            | 0.23 | 0.07 | 32                   | 0.26 | 0.06 |
| Lateral lane position (m)     | 33       | 0.22 | 0.01 | 33                     | 0.16 | 0.08 | 32                            | 0.15 | 0.07 | 32                   | 0.18 | 0.08 |
| Average speed (km/h)          | 33       | 43   | 3.2  | 33                     | 44   | 3.08 | 33                            | 44   | 3.4  | 33                   | 43   | 3.6  |
| Speed limit exceedance (%)    | 33       | 27   | 11   | 33                     | 28   | 9.7  | 33                            | 23   | 11   | 33                   | 21   | 9.5  |
| Mental workload               | 33       | 1.4  | 0.7  | 33                     | 0.9  | 0.6  | 33                            | 1.4  | 0.8  | 33                   | 2.0  | 1.0  |
| - Attention                   | 33       | 2.2  | 1.3  | 33                     | 1.7  | 1.0  | 33                            | 2.2  | 1.2  | 33                   | 2.7  | 1.5  |
| - Visual                      | 33       | 2.1  | 1.3  | 33                     | 1.7  | 0.9  | 33                            | 2.1  | 1.2  | 33                   | 2.3  | 1.7  |
| - Auditory                    | 33       | 0.5  | 0.8  | 32                     | 0.5  | 0.8  | 32                            | 0.4  | 0.6  | 32                   | 0.7  | 0.8  |
| - Stress                      | 33       | 1.6  | 1.3  | 32                     | 0.8  | 0.8  | 33                            | 1.7  | 1.2  | 33                   | 2.6  | 1.4  |
| - Temporal                    | 33       | 0.7  | 1.0  | 32                     | 0.4  | 0.6  | 33                            | 0.7  | 0.9  | 33                   | 1.9  | 1.5  |
| - Interference                | 33       | 1.1  | 1.2  | 33                     | 1.0  | 1.1  | 33                            | 1.7  | 1.3  | 33                   | 2.2  | 1.7  |
| Perceived Safety              | 33       | 5.7  | 1.0  | 33                     | 5.4  | 1.1  | 33                            | 5.2  | 1.2  | 33                   | 2.3  | 1.0  |
| User Experience               | 33       | 0.01 | 0.7  | 33                     | 1.2  | 0.6  | 33                            | 1.4  | 0.7  | 33                   | 0.5  | 0.7  |
| - Pragmatic                   | 33       | 1.3  | 1.0  | 33                     | 1.7  | 0.9  | 33                            | 1.5  | 0.9  | 33                   | -0.5 | 1.1  |
| - Hedonic                     | 33       | -1.3 | 0.9  | 33                     | 0.6  | 0.6  | 33                            | 1.3  | 0.7  | 33                   | 1.5  | 0.7  |
| Ranking                       |          |      |      |                        |      |      |                               |      |      |                      |      |      |
| - Driving Performance         | 33       | 2.3  | 1.1  | 33                     | 3.1  | 0.7  | 33                            | 3.1  | 1.0  | 33                   | 1.5  | 0.8  |
| - Safety                      | 33       | 2.8  | 0.9  | 33                     | 3.1  | 0.7  | 33                            | 2.9  | 1.0  | 32                   | 1.1  | 0.3  |
| - Mental workload             | 33       | 2.5  | 1.0  | 33                     | 2.2  | 1.1  | 33                            | 2.3  | 1.0  | 33                   | 3.1  | 1.3  |

Note. n = number of participants, M = mean, SD = standard deviation. Number of participants other than 33 are indicated in italics.